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New paleomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan

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ABSTRACT

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We report new results from our paleomagnetic studies of the Inner Mongolia autonomous region in northeastern China and the Xinjiang autonomous region in northwestern China. Primary magnetizations are isolated in Tertiary, Cretaceous, Late Jurassic, and Late Permian rocks, and secondary components of magnetization of Late Jurassic and Recent ages are seen in strata of Inner Mongolia. Western Junggar Basin, and part of the Tarim block, suffered widespread complete tectono-thermal remagnetization at about 280 Ma (Permian). The paleomagnetic evidence suggests that the sampled areas in Inner Mongolia have been part of the North China Block at least since Late Permian time, and the Junggar block collided with the Tarim block by Late Permian time. Our own paleomagnetic results together with available data from other parts of eastern Eurasia, suggest a tectonic model in which Kazakhstan, Europe, and Siberia collided with each other and formed the Europe–Siberia–Kazakhstan plate (EUR–SIB–KAZ) in Late Permian time. To the east, Mongolia (= Outer Mongolia), Inner Mongolia, and the North China Block were also wedged together forming the Mongolia–North China plate (MON–NC) by this time. These two composite continental plates were not a stable unit in the Late Permian. The geometrical relationship of the paleomagnetic poles for these two plates suggests that collision initially occurred in the Late Permian near the western end (50° N, 75° E) of the Southern Siberian Fold Belt and progressed eastward as the MON–NC plate rotated and clockwise about 117° relative to the EUR–SIB–KAZ plate during the Early Mesozoic. Suturing and rotation were probably complete by the Late Jurassic. Significant relative motion between the Tarim and Siberia blocks continued from Early Triassic to Late Jurassic, as Tarim moved northeastward 1400 km relative to Siberia.

Introduction

The eastern part of Eurasia is a composite continent formed by the amalgamation of several continental and microcontinental blocks (McElhinny et al., 1981; Klimetz, 1983; Il'in, 1986; Maruyama et al., 1989). Several fold belts that separate these continental blocks are considered to be of Paleozoic and Early Mesozoic ages and are interpreted as the sites of convergent plate margins (Klimetz, 1983; Li et al., 1984; Zhang et al., 1984; Yang and Yang, 1985; Zonenshain et al., 1985). Although the geology of the fold belts between these blocks has the potential of constraining the times of collisions more tightly than the geology of the blocks themselves, the complexity of these fold belts has led to widely different views of the tectonic history of the region in terms of the style of convergence and timing of amalgamation of individual continental blocks. For example, geological estimates on the timing of the collision along the east–west trending Qinling Fold Belt to the south (Fig. 1a), which separates the North China Block (NCB) and the South China Block (SCB), range from Devonian to Early Jurassic (Klimetz, 1983; Li et al., 1984; Zhang et al., 1984; Mattauer et al., 1985; Sengör, 1985; Il'in, 1986; Yang et al., 1986; Hsü et al., 1987; Watson et al., 1987; Liu and Hao, 1989).

The situation is not much clearer either with respect to the NCB and the Tarim block Many Chinese geologists believe that the two blocks

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were welded together in the Late Proterozoic (Huang, 1978; Li et al., 1984; Zhang et al., 1984; Ren et al, 1987), forming a stable coherent continent/plate. On the other hand, Ji and Coney (1985) and Coleman (1989) envision a complex process of amalgamation of these two blocks with each other and with blocks to the north during the Late Paleozoic.

To the north, most geological investigators have suggested that the Chinese blocks (including the North China Block to the east and the Junggar and Tarim blocks to the west) accreted to Siberia (SIB) and the adjacent Soviet blocks no late than Late Permian time along the Central Asian Fold Belt (also called the Inner Mongolia–Greater Hinggan Fold Belt, the Tianshan–Hegan–Xar Moron fold zone, and the Variscan (late Paleozoic) Fold Belt (Huang, 1978; Ren et al., 1980, 1987; Wang, 1981; Li and Tang, 1983; Li et al., 1984; Zhang et al., 1984; Chen and Liang, 1985; Ji and Coney, 1985; Yang and Yang, 1985; Wang and Liu, 1986; Yang et al., 1986; Coleman, 1989; Feng et al., 1989; Maruyama et al., 1989). For instance, Ren et al. (1980, 1987) concluded that the various blocks within China were connected to Siberia during the Hercynian orogenic phases. Using terminology from plate tectonic theory, Li and Tang (1983), Yang and Yang (1985), and Yang et al. (1986) concluded that the sea trough between the Siberian and Chinese blocks was closed in the Late Paleozoic. Similar conclusions were also reached by analyzing paleobiogeographic data (Wang, 1981; Wu, 1981; Wang and Liu, 1986). However, ever since the pioneering study by McElhinny et al. (1981), paleomagnetic data have strongly contradicted these geological assertions for pre-Late Permian suturing of China to Siberia (see below). Moreover, there are a few geological studies as well that support a post-Permian date of amalgamation. Klimetz (1983) argued that collision between Chinese blocks and Siberia did not occur until Late Jurassic to Early Cretaceous time, based mainly on his stratigraphic summary and analysis for eastern China,

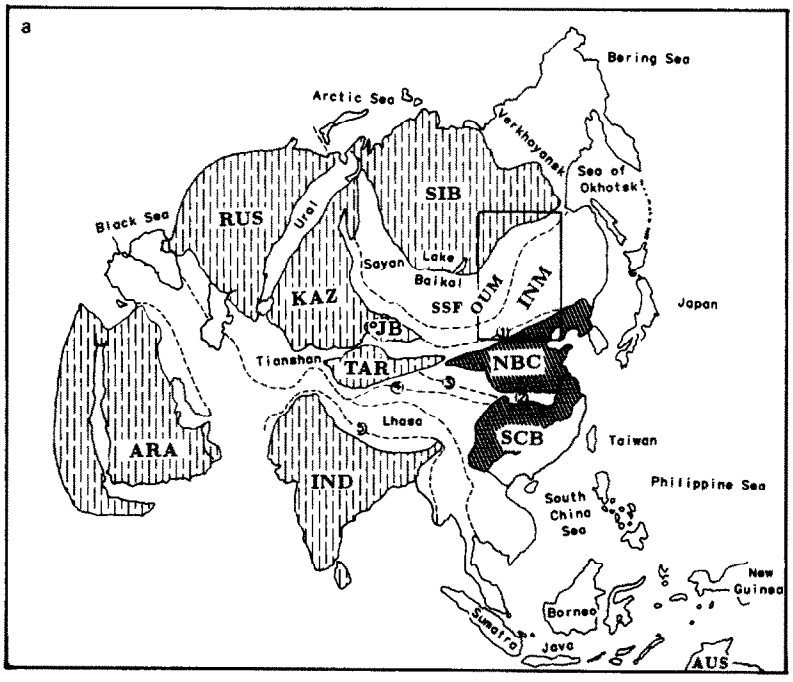


Fig. 1a. Sketch map of present location of continental blocks of eastern Asia. Abbreviations: *SIB* = Siberia; *RUS* = Russia; *KAZ* = Kazakhstan; *JB* = Junggar Basin; *SSF* = Southern Siberian Fold Belt; *OUM* = Outer Mongolia; *INM* = Inner Mongolia; *NCB* = North China Block; *SCB* = South China Block; *TAR* = Tarim; *IND* = India; *ARA* = Arabian Shield; *AUS* = Australia. Numbers: *1* = Inner Mongolia Greater Hinggan Feld Belt; *2* = Qinling Fold Belt; *2* = Nanshan Fold Belt; *4* = Kunlun Fold Belt; *5* = Himalayan Fold Belt. The rectangle and small circle outline the sampling areas of Inner Mongolia (Fig. 1b) and west Junggar (Fig. 1c), respectively.

Mongolia, and the southern Soviet Far East. Zonenshain et al. (1985), from Paleozoic ocean reconstructions, concluded that Siberia and China collided in the Early Triassic. Huang (1986) and Nie et al. (1989) also favor a final collision during the Mesozoic, based on recently discovered fossil plants and climatic indicators in Inner Mongolia.

Paleomagnetism has proved to be a powerful tool in studying the tectonic evolution of continental blocks and in testing alternative geological hypotheses of accretionary history, both in China and elsewhere. For example, paleomagnetic results for Late Permian rocks of both the NCB and SCB have shed some light on the controversy of when and how the two blocks collided and amalgamated, and summaries of these results and their implications have been discussed previously (Zhao and Coe, 1987, 1989). Very briefly, the paleomagnetic evidence supports the hypothesis that the NCB and SCB were joined together into a stable unit sometime between Early Triassic and Cretaceous time, with collision initially occurring toward the eastern end of the NCB–SCB boundary

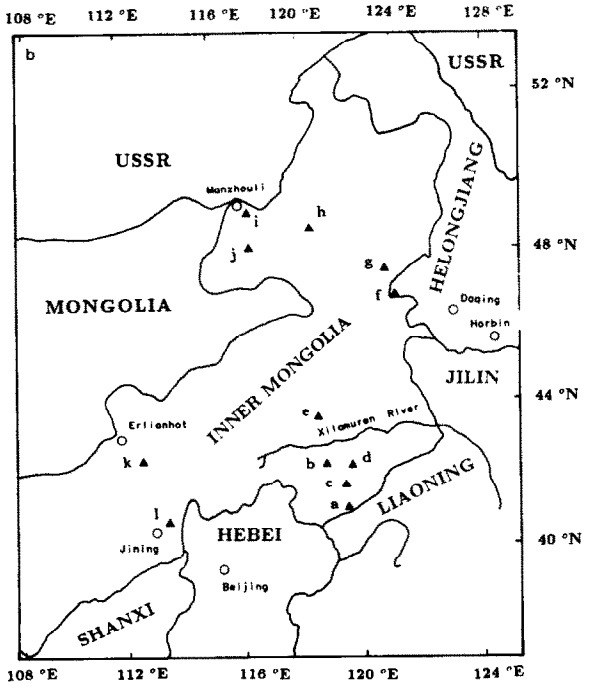


Fig. 1b. Paleomagnetic sampling localities of this study in Inner Mongolia. Characters correspond to those in Table 1. Triangles represent paleomagnetic sampling areas.

and suturing progressing westward as the SCB rotated clockwise relative to the NCB. With regard to the amalgamation of western China, all Late Permian paleomagnetic poles for the Tarim block are very different from those of the NCB, showing that the Tarim block and the NCB were not likely joined in Permian time (Bai et al., 1987; Li et al., 1988; McFadden et al., 1988; Sharps et al., 1989).

As alluded to above, Late Permian to Middle Triassic paleomagnetic data from China and Siberia imply that both the NCB and SCB lay several thousand kilometers further south of Siberia than at present (McElhinny et al., 1981; Opdyke et al., 1986; Zhao and Coe, 1987; Cheng et al., 1988; Tan, 1988; Wu, 1988; Lin, 1989). Paleomagnetic data from more northerly parts of China have been urgently needed to help resolve the problem of where and when the displacement between China and Siberia took place. Recently we have sampled extensively for paleomagnetic studies in both the Inner Mongolia and Xinjiang autonomous regions, which are situated within the Central Asian Fold Belt in northern China (Fig. 1). We reported preliminary results from these studies at several meetings (Coe et al., 1988; Zhao et al., 1988), and gave a tentative interpretation of them in a recent article that dealt mainly with the amalgamation of the NCB and SCB (Zhao and Coe, 1989). There we proposed a simple tectonic model in which initial collision of the Chinese and Siberian blocks occurred in the Late Permian near the western end of the east–west trending South Siberian Fold Belt (Fig. 1) and progressed eastward as the Chinese blocks rotated anticlockwise relative to Siberia during the Mesozoic.

In this paper we present the detailed account of our own paleomagnetic results in the eastern (Inner Mongolia) and western (Junggar) parts of the Central Asian Fold Belt and gather other relevant paleomagnetic data from various blocks of eastern Eurasia, some of which are not currently available in the western literature. By evaluating and comparing the available paleomagnetic data, we refine our tectonic model for the collision and relative rotation of these blocks and illustrate semiquantitatively the amounts of rotation involved at some key points in geologic time.

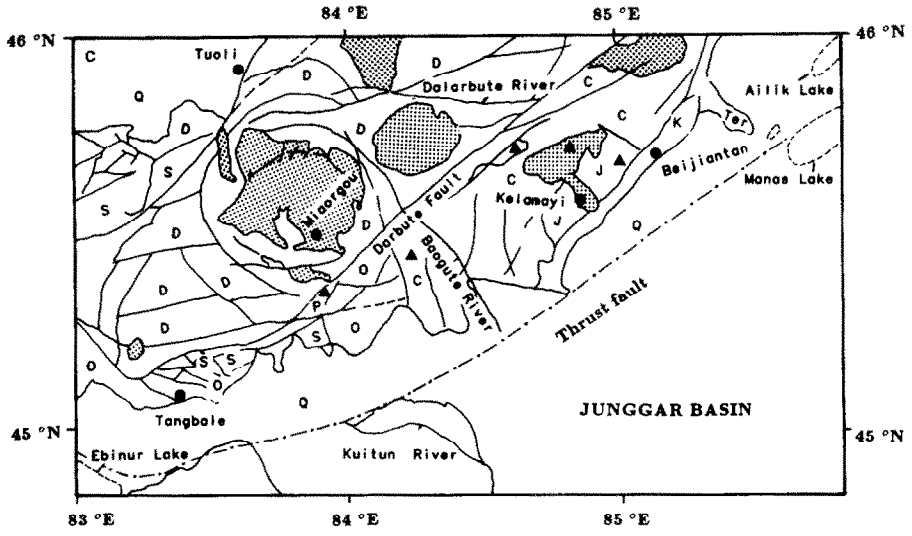


Fig. 1c. Details of paleomagnetic sampling areas (triangles) in west Junggar. Age abbreviations; O = Ordovician; S = Silurian; D = Devonian; C = Carboniferous; P = Permain; J = Jurassic; Ter = Tertiary.

Geologic setting and paleomagnetic sampling

The territories of the Inner Mongolia and Xinjiang autonomous regions are situated at the junction of convergence between four major Chinese and Soviet blocks. On the north is the Siberian block, to the south are the North China and the Tarim blocks, and to the west is the Kazakhstan block (Fig. 1a).

Inner Mongolia

The geologic setting of the areas sampled in Inner Mongolia is complex and is very different from those of the NCB (Nei Monggol Regional Stratigraphy Tables Compilation Team, 1978). Briefly, the Late Proterozoic to Early Paleozoic strata are mainly clastics with some shallow water carbonates, evaporites, and volcanic rocks. These rocks have all been metamorphosed to various degrees. The ages of the two formations containing these strata (Zhaertay Group and Baiyun'ebo Group) have long been debated, with estimates ranging from Late Proterozoic to Silurian (Yang and Yang, 1985; Wang and Liu, 1986). For this reason, and because of significant metamorphism, we did not sample rocks that are of pre-Carboniferous age. Carboniferous rocks are mainly sedimentary, including limestones, dolomites, redbeds, and shales. The Permian is represented by both sedimentary and volcanic rocks, consisting of red sandstones, dolomites, metasediments, and andesitic lava flows. Mesozoic to Cenozoic sequences consist of oil-and-gas-bearing sediments of mainly continental facies, and include basaltic extrusive rocks and granodiorite intrusions. Jurassic rocks are all volcanic, mainly andesitic and tuff flows, and Cretaceous formations that we sampled are tuffaceous sediments and volcaniclastic rocks. Miocene basaltic lava flows are also present. Unfortunately, no rocks that are definitely of Triassic age have been found in this region, probably due to regional uplift during this time (Quan, 1983). Major regional folding and deformation of the strata in Inner Mongolia occurred during the Late Hercynian, Indosinian, and Yanshanian orogenies, which correspond to the Late Permian, Late Triassic to Early Jurassic, and Jura-Cretaceous, respectively. The Yanshanian orogeny was mainly confined to northeastern Inner Mongolia and was the most intensive and extensive in terms of igneous activity and regional deformation. Igneous rocks of this orogeny yield K–Ar isotopic age values between 140 and 169 Ma (Quan, 9183; T.D. Wang, 1986), corresponding to Late Jurassic time (Harland et al., 1982; Palmer, 1983; Snelling, 1987).

We made paleomagnetic sampling expeditions in the summers of 1987 and 1988 and collected a total of 920 individual oriented samples at 111

TABLE 1

Paleomagnetic results for rock units sampled in Inner Mongolia

Study site No. [a]	Area (°N, °E)	Rock	Age	Strike/ dip	Polarity ratio R/N	T/H range	Comp.	Statistical analysis								
								N/N_0	directions				poles			
									D	I	α_{95}	k	long.	lat.	A_{95}	K
a. Pingzhuang																
A1	42.0, 119.2	basalt	T	198/20	1/7	100–200	1	8/8	39.5	56.6	16.3	12.5	204.4	62.3	19.4	9.1
					0/8	350–525	2	8/8	63.2	48.5	3.7	229.7	201.4	38.4	3.7	229.7
A2	42.0, 119.2	basalt	T	0/0	0/6	100–325	1	6/8	343.7	−20.0	13.1	27.0	318.7	35.5	10.9	39.0
					0/8	300–550	2	8/8	350.9	46.7	13.1	18.7	333.5	74.1	16.5	12.2
A3	42.0, 199.2	basalt	T	337/23	0/7	375–550	2	7/7	327.8	55.9	25.9	6.4	26.0	67.7	31.5	4.6
A4	42.0, 119.2	basalt	T	21/27	0/5	400–500	2	5/5	338.9	35.1	17.5	20.0	344.3	61.6	17.0	21.2
A5	42.0, 119.2	basalt	T	50/56	10/0	475–630	2	10/10	156.2	−13.7	20.2	6.7	157.1	−49.9	14.2	12.5
A6	42.0, 119.2	basalt	T	37/46	11/0	475–650	2	11/11	157.2	−14.1	30.3	3.2	154.6	−48.8	26.4	3.9
A7	42.0, 119.2	tuff. sed.	K1	265/23	0/7	300–450	2	7/7	43.0	76.6	7.9	59.8	150.8	56.4	14.0	19.6
A8	42.0, 119.2	tuff. sed.	K1	260/12	3/6	100–350	2	9/9	2.9	64.9	12.0	19.4	122.6	82.2	15.3	12.2
A9	42.0, 119.2	tuff. sed.	K1	260/12	0/8	100–400	2	8/8	9.2	75.2	5.8	93.5	131.1	68.6	10.0	31.7
A10	42.0, 119.2	tuff. sed.	K1	260/12	0/10	150–400	2	10/10	9.6	66.7	5.5	78.3	158.0	79.4	8.5	33.3
b. Wudan																
B1	42.7, 118.9	redbeds	P2	305/19	0/6	300–550	1	6/7	357.6	56.8	6.0	126.1	322.8	84.4	7.7	75.7
					0/6	550–690	2	6/7	162.0	−22.2	7.7	76.5	151.1	−55.2	6.2	119.4
B2 [b]	42.7, 118.9	redbeds	P2	295/21	7/0	630–730	2	7/7	183.7	−35.2	24.6	7.6	110.1	−66.5	12.1 [c]	16.7 [c]
B3	42.7, 118.9	sandstone	P2	87/7	0/4	350–475	1	4/5	12.9	66.4	10.9	72.0	164.9	79.1	14.9	39.0
B4	42.7, 118.9	andesite	P2	300/20	0/6	300–500	1	6/6	8.9	66.9	3.9	302.6	156.8	80.6	5.7	140.1
					6/0	500–610	2	6/6	156.8	−33.7	3.2	448.9	165.1	−58.8	3.2	440.1
B5	42.7, 118.9	andesite	P2	295/21	0/5	300–500	1	5/6	2.1	66.9	4.5	284.4	130.3	82.9	6.9	122.4
					6/0	550–600	2	6/6	148.7	−26.1	3.5	368.4	171.3	−50.5	3.2	446.4
B6	42.7, 118.9	andesite	P2	300/20	0/5	300–500	1	5/5	5.1	66.5	8.2	88.8	145.2	82.6	12.9	36.1
					5/0	550–600	2	5/5	164.9	−27.1	8.7	78.8	147.9	−58.9	8.9	74.8
B7	42.7, 118.9	andesite	P2	300/20	0/4	550–630	2	4/4	339.9	27.4	12.8	70.1	336.5	57.0	19.2 [c]	7.0 [c]
c. Aoji																
C1	42.2, 119.3	sandstone	P1	303/84	0/5	100–625	1	5/6	352.3	74.3	9.9	60.9	110.8	70.6	16.2	23.2
C2	42.3, 120.4	SS and shale	C	48/54	2/4	600–680	2	6/6	97.5	62.3	11.1	37.5	171.6	23.4	16.9	16.7
C3	42.3, 120.4	sandy shale	C	56/55	0/8	100–450	1	8/8	46.3	68.9	4.1	187.3	175.5	57.6	6.2	80.0
C4	42.4, 120.4	redbeds	C	66/42	2/5	500–680	1	7/7	24.8	58.2	4.6	172.2	212.4	71.1	6.1	99.6
C5	42.3, 120.3	redbeds	C3	69/49	0/6	300–500	1	6/10	10.8	62.9	12.2	31.0	184.7	81.2	16.1	18.4
					5/5	570–690	2	10/10	36.2	62.0	6.4	58.4	194.4	63.2	8.2	34.1

TABLE 1 (continued)

Study site No. [a]	Area (°N, °E)	Rock	Age	Strike/ dip	Polarity ratio R/N	T/H range	Comp.	Statistical analysis							
								N/N_0	directions				poles			
									D	I	α_{95}	k	long.	lat.	A_{95}	K
C6	42.3, 119.9	limestone	P1	251/58	0/5	100–400	1	5/11	48.7	69.4	34.8	5.8	165.9	56.2	55.9	2.8
C7	42.3, 119.9	limestone	P1	251/58	0/8	375–400	1	8/8	7.4	66.4	14.4	15.8	145.2	78.4	20.6	8.2
C8	42.3, 119.9	limestone	P1	251/58	0/5	375–400	1	5/6	10.2	51.5	8.5	81.7	257.7	77.4	9.5	66.3
d. Shizhangfang																
D1	43.0, 118.9	granite	J2-3	0/0	0/5	100–300	1	5/6	8.8	65.9	4.5	291.6	165.1	82.0	7.0	122.1
					5/0	400–600	2	5/6	212.3	−51.8	2.9	702.8	40.5	−62.6	3.0	655.7
D2 [b]	43.0, 118.9	andesite	P2	200/22	0/4	100–200	1	4/4	1.6	69.7	7.6	147.3	124.2	78.9	11.5	64.8
					5/0	500–670	2	5/5	139.5	−33.8	19.8	15.9	186.8	−48.1	20.7	14.7
D3 [b]	43.0, 118.9	andesite	P2	200/22	0/2	50–150	1	2/7	0.9	57.3		7.6	147.3	89.2		6.0
					3/0	550–670	2	3/7	175.9	−47.1		8.1	123.8	−77.0		7.1
D4	43.0, 118.9	andesite	P2	191/34	0/8	100–300	1	8/8	1.5	61.3	2.1	693.1	234.2	88.8	2.6	444.5
					1/7	500–600	2	8/8	352.5	48.1	5.0	125.7	325.1	75.2	5.2	112.6
D5	43.0, 118.9	andesite	P2	191/34	0/6	100–300	1	6/6	354.1	66.8	2.6	688.7	88.2	82.4	4.1	265.1
					0/6	350–600	2	6/6	347.1	47.1	4.6	210.3	338.7	72.2	4.3	246.9
D6	43.0, 118.9	andesite	P2	191/34	0/4	100–300	1	4/6	12.1	60.4	12.5	55.0	210.8	81.7	14.9	38.8
					0/6	300–600	2	6/6	351.8	53.6	10.6	40.9	341.8	79.5	13.0	27.4
e. Balinzuo Qi																
E1	43.9, 118.9	andesite	P1	121/39	0/4	100–400	1	4/7	2.8	71.8	24.8	14.7	116.3	74.4	38.6	6.6
					0/7	500–600	2	7/7	96.1	77.8	9.7	39.3	147.1	37.6	17.8	12.5
E2	43.9, 118.9	andesite	P1	121/39	0/6	100–200	1	6/7	18.6	40.7	14.0	23.9	255.5	64.7	15.4	19.9
					0/7	450–625	2	7/7	63.1	61.9	7.0	74.7	186.1	45.8	9.7	39.8
E3	43.8, 119.0	sandstone	P2	193/65	0/10	100–500	1	10/10	350.5	74.3	3.1	251.8	103.8	72.0	5.2	87.5
					9/0	500–685	2	9/10	241.4	−29.1	7.1	54.0	37.3	−33.3	6.9	56.2
E4	43.8, 119.0	metasediment	P2	193/63	0/7	200–450	1	7/10	46.7	58.0	7.0	75.4	199.9	54.8	8.5	51.6
					6/0	500–700	2	6/10	247.4	−22.8	22.9	9.5	35.1	−24.7	19.6	12.6
E5	43.8, 119.0	metasediment	P2	102/65	0/5	300–500	1	5/5	16.6	71.2	7.8	97.2	154.1	73.4	12.1	40.7
E6	43.9, 118.9	andesite	P1	60/30	2/6	400–585	1	8/8	306.1	73.7	17.6	10.9	81.3	54.3	26.2	5.4
E7	44.0, 118.9	andesite	P1	60/30	5/0	350–550	1	5/6	156.2	−41.5	17.6	19.8	173.5	−61.8	18.4	18.2
					0/6	550–625	2	6/6	49.9	82.3	6.4	109.6	137.5	52.5	11.9	32.7
E8	43.9, 120.1	limestone	C	134/72	0/5	100–300	1	5/8	351.1	65.7	10.9	50.2	76.2	81.9	13.6	32.4
E9	43.9, 120.1	limestone	C	221/88	0/9	100–400	1	9/11	9.7	74.4	5.6	84.9	135.2	71.3	9.5	30.4
					11/0	450–600	2	11/11	162.4	−64.2	6.6	49.3	229.6	−77.6	9.5	24.1
E10	43.9, 120.1	limestone	C	221/88	0/5	100–450	1	5/7	337.8	57.9	19.0	17.2	28.6	75.2	23.9	11.2
					6/0	450–575	2	6/7	163.5	−59.6	7.9	73.2	201.6	−78.1	9.6	49.8

E11	43.9, 120.1	limestone	C	134/72	0/4	150–400	1	4/12	14.9	68.7	8.7	112.4	162.6	76.8	12.3	56.9
					12/0	450–600	2	12/12	142.4	16.2	9.8	20.7	163.2	−27.3	7.4	35.2
E12	43.9, 119.5	tuff	J3	270/30	0/9	200–500	1	9/9	6.0	68.1	2.4	453.1	145.9	81.6	3.8	187.0
					7/0	525–640	2	7/9	246.0	−79.0	7.8	60.2	328.9	−48.5	14.6	18.1
E13	43.8, 119.0	metasediment	P2	103/65	0/5	100–450	1	5/5	352.5	70.6	8.0	92.8	97.2	77.5	13.6	32.7
E14	43.9, 118.9	andesite	P1	121/39	0/10	475–650	2	10/10	320.5	13.1	40.9	2.4	349.1	39.3	37.8	2.6
E15	43.9, 118.9	andesite	P1	60/30	4/0	500–650	2	4/8	63.3	−36.4	6.5	203.7	241.9	3.6	4.9	347.2
f. Hua-an																
F1	47.0, 123.1	andesite	J3	261/35	0/6	600–665	2	6/6	73.2	66.1	3.7	328.5	182.7	42.6	5.5	148.0
F2	47.0, 123.1	andesite	J3	261/35	0/7	625–670	2	7/7	66.6	69.0	3.0	400.7	179.9	48.3	4.9	152.1
F3	47.0, 123.1	andesite	J3	261/35	0/5	650–720	2	5/5	61.1	67.0	4.4	297.8	185.6	50.6	6.5	139.3
F4	47.0, 123.1	sandstone	P1	110/22	0/10	500–600	1	10/10	11.2	74.4	5.0	97.7	142.1	74.8	8.5	33.1
F5	47.0, 123.1	sandstone	P1	122/28	0/5	475–600	1	5/6	5.9	71.2	5.4	201.4	142.6	80.4	8.9	75.7
F6	47.0, 123.1	andesite	J3	224/15	0/7	575–625	2	7/7	27.0	68.3	5.3	131.0	187.5	72.0	8.3	54.4
F7	47.0, 123.1	andesite	J3	304/15	0/11	550–625	2	11/11	351.0	66.5	4.6	101.3	57.2	83.7	7.1	42.7
F8	47.0, 123.1	andesite	J3	304/15	0/8	500–625	2	8/8	347.4	68.6	5.7	97.0	71.1	80.6	9.3	36.8
F9	47.0, 123.1	andesite	J3	304/15	0/12	450–675	2	12/13	13.5	66.3	4.4	99.4	194.3	80.8	6.5	46.1
g. Zalantum																
G1	48.0, 122.8	andesite	J3	230/25	0/6	100–400	1	6/7	3.9	70.3	6.2	116.1	141.5	82.7	9.7	48.3
					7/0	450–620	2	7/7	262.1	−66.1	5.6	117.9	358.7	−38.2	8.0	58.6
G2	48.2, 122.6	sandstone	C3	313/36	0/3	100–200	1	3/5	353.5	60.0	16.3	58.2	342.3	82.2	20.1	38.5
					3/0	300–400	2	3/5	197.2	−40.8	6.1	406.5	87.8	−61.7	7.0	314.8
G3	48.2, 122.6	andesite	J3(?)	117/75	0/6	100–300	1	6/7	357.6	71.6	5.7	140.9	114.4	81.3	9.0	56.8
					2/0	575–660	2	2/7	248.3	−38.7	–	24.7	33.3	−31.2	–	26.1
G4	48.2, 122.6	andesite	P	20/57	0/4	100–300	1	4/7	1.6	72.1	7.8	139.6	127.7	80.5	12.6	54.2
					4/0	400–525	2	4/7	241.4	−41.7	32.9	8.8	35.8	−38.9	34.3	8.2
h. Hailaer																
H1	49.2, 119.8	limestone	C2	20/24	0/8	100–300	1	8/8	357.7	69.2	5.5	101.5	98.8	85.7	9.0	39.2
					8/0	300–550	2	8/8	178.9	−58.2	1.8	989.5	124.6	−79.7	2.3	561.3
H2	49.2, 119.8	limestone	C2	15/20	9/0	560–645	2	9/9	169.5	−56.1	4.5	132.1	156.3	−75.5	5.7	82.4
H3	49.2, 119.8	limestone	C2	20/24	0/4	100–200	1	4/6	14.8	60.4	11.5	64.8	240.7	76.9	17.3	29.3
					6/0	300–600	2	6/6	177.8	−58.8	8.7	59.8	134.5	−80.6	11.4	35.5
H4	49.2, 119.8	limestone	C2	21/22	0/5	100–300	1	5/8	8.1	60.5	10.6	53.5	259.0	80.8	15.3	25.9
					8/0	300–625	2	8/8	188.5	−49.8	5.5	110.3	97.2	−70.6	5.9	87.9
i. Manzhouli																
I1	49.5, 117.5	tuff	J3	98/12.5	6/0	500–625	2	6/6	214.8	−37.0	2.8	573.7	61.2	−50.2	2.7	594.7
I2	4.95, 117.5	tuff	J3	98/12.5	7/0	550–620	2	7/7	211.3	−41.1	6.1	98.0	62.7	−54.6	6.3	92.2
I3	49.5, 117.5	tuff	J3	115/15	9/0	525–640	2	9/9	208.9	−42.6	5.3	90.7	64.5	−56.8	5.4	91.3
I4	49.5, 117.5	tuff	J3	120/19	6/0	525–640	2	6/6	214.7	−30.3	2.2	897.4	64.8	−46.5	2.110	10.8
I5	49.5, 117.5	tuff	J3	107/15	6/0	525–640	2	6/6	217.4	−37.8	9.8	47.5	57.5	−49.4	9.1	54.9
I6	49.5, 117.5	tuff	J3	125/16	8/0	525–670	2	8/8	220.1	−40.9	5.0	124.6	52.4	−49.4	5.3	108.6

TABLE 1 (continued)

Study site No. [a]	Area (°N, °E)	Rock	Age	Strike/ dip	Polarity ratio R/N	T/H range	Comp.	Statistical analysis								
								N/N_0	directions				poles			
									D	I	α_{95}	k	long.	lat.	A_{95}	K
I7	49.5, 117.4	tuff	J3	153/14	6/0	575–670	2	6/6	219.9	−41.8	5.1	175.0	52.0	−50.0	5.3	161.8
I8	49.5, 117.5	tuff	J3	108/17	6/0	600–670	2	6/6	218.8	−54.1	7.4	83.2	40.3	−58.2	9.9	46.6
I9	49.4, 117.6	andesite	J3	11/8	0/5	100–400	1	5/7	340.2	67.5	15.6	24.9	45.5	77.6	24.4	10.8
					0/7	575–625	2	7/7	7.3	59.9	2.2	731.4	264.0	80.0	3.1	389.3
I10	49.4, 117.6	andesite	J3	0/9	0/7	400–625	2	7/7	10.3	59.2	2.3	692.7	255.8	78.2	2.8	466.5
I11	49.4, 117.6	andesite	J3	346/7	6/0	500–625	1	6/7	236.2	−69.1	13.6	25.1	356.9	−55.3	21.5	10.7
					0/7	600–700	2	7/7	19.9	54.7	3.1	380.7	244.4	69.7	4.0	226.1
I12	49.4, 117.6	andesite	J3	2/8	0/10	525–650	2	10/10	7.0	59.8	3.8	164.4	263.7	80.1	5.1	89.8
I13	49.7, 117.7	tuffite	J3	114/15	0/3	50 Oe	1	3/9	3.7	52.1	9.5	169.8	287.7	73.0	10.4	140.8
					9/0	575–665	2	9/9	204.0	−34.2	4.0	163.6	77.1	−53.7	3.7	190.8
I14	49.7, 117.7	tuffite	J3	114/15	7/0	600–675	2	7/7	206.9	−42.4	5.9	103.9	67.8	−57.6	5.2	134.9
j. Dalai Lake																
J1	48.9, 117.0	volc. rock	K1	339/7	0/9	100–300	1	9/11	4.2	67.8	4.6	128.4	165.5	86.3	6.9	56.7
					11/0	450–600	2	11/11	174.5	−65.2	5.5	69.3	198.7	−86.1	6.8	46.6
J2	48.9, 117.0	zeolite	K1	357/9	0/7	100–400	1	7/9	21.6	71.0	6.3	93.5	171.3	75.0	10.3	35.0
					9/0	500–635	2	9/9	164.8	−60.6	4.8	117.8	178.2	−77.3	6.7	60.4
k. Minganaobao																
K1	43.1, 113.2	limestone	C2	50/39	5/5	200–500	1	10/11	359.4	55.4	7.0	49.0	295.4	83.9	8.0	37.1
					11/0	400–650	2	11/11	230.2	−41.0	7.7	35.7	31.2	−44.7	8.1	32.5
K2	43.1, 113.2	limestone	C2	50/72	0/7	100–350	1	7/7	352.7	65.0	5.3	131.6	66.5	83.6	7.9	59.8
					7/0	425–585	2	7/7	220.9	−35.3	9.6	40.7	44.2	−48.8	9.1	44.5
K3	43.1, 113.2	limestone	C2	50/60	0/9	100–350	1	9/9	348.6	65.7	6.3	68.7	58.8	80.4	9.6	29.8
					9/0	350–600	2	9/9	195.5	−50.9	8.9	34.6	62.5	−74.2	10.7	24.0
K4	43.1, 113.2	limestone	C2	50/39	0/7	100–300	1	7/8	344.9	63.9	7.6	64.8	42.7	78.9	11.1	30.6
					8/0	400–625	2	8/8	228.6	−35.1	10.9	26.7	37.5	−43.9	9.4	35.4
K5	43.1, 113.2	sandstone	C2	50/39	0/7	100–300	1	7/9	346.0	64.8	8.0	57.6	53.2	79.1	11.7	27.8
					6/0	350–675	2	6/9	243.4	−39.1	16.2	18.0	22.9	−34.7	15.9	18.8
K6	43.1, 113,2	mudstone	P1	13/67	0/14	200–400	1	14/14	358.5	60.1	4.7	73.6	330.4	87.8	6.8	35.3
					0/11	300–600	2	11/11	46.9	41.6	11.4	17.1	214.5	47.8	13.5	12.4
K7	43.0, 113.2	sandstone	P1	8/64	10/0	500–600	2	10/15	238.8	22.8	9.9	24.9	54.3	−13.1	8.8	31.4
K8	43.0, 113.2	sandstone	P1	13/67	0/9	100–400	1	9/9	352.9	56.8	15.0	12.7	13.2	85.4	17.9	9.3
					3/0	300–550	2	3/9	255.1	27.9	48.3	7.6	43.7	0.0	52.9	6.5
K9	43.0, 113.2	sandstone	P1	13/67	0/5	100–400	1	5/7	359.5	61.8	12.7	37.3	91.5	89.2	17.3	20.6
					2/2	300–625	2	4/7	0.7	38.2	27.6	12.0	289.5	69.5	26.4	13.0

l. Jining																
L1	40.9, 113.2	basalts	T	0/0	0/7	575–675	2	7/7	345.6	55.3	4.9	155.0	4.0	77.7	6.4	89.6
L2	40.9, 113.2	basalts	T	0/0	0/12	575–625	2	12/12	358.0	53.3	2.0	352.1	305.9	82.8	2.6	212.1
L3	40.9, 113.2	basalts	T	0/0	0/6	585–675	2	6/6	354.0	48.9	5.4	153.9	320.1	78.1	5.1	176.6
L4	40.9, 113.2	basalts	T	0/0	0/5	525–675	2	5/5	357.5	54.6	8.1	89.5	314.1	84.6	9.2	70.5
L5	40.9, 113.2	basalts	T	0/0	5/0	525–585	2	5/5	180.3	−68.4	4.8	254.0	293.3	−79.1	7.7	98.5
l. Jining																
L6	40.9, 113.2	basalts	T	0.0	5/0	575–625	2	5/5	162.6	−76.2	8.5	81.3	275.3	−64.4	15.2	26.3
L7	40.9, 113.2	basalts	T	0/0	6/0	615–675	2	6/6	173.8	−75.9	6.5	107.8	286.4	−66.8	11.2	36.8

Rock: rock type.

Age: approximate age of rocks studied, based mainly on palentological and geological evidence.

T = Tertiary, J = Jurassic, P = Permian, C = Carboniferous; 1 = Early, 2 = Middle, 3 = Late.

Polarity ratio: number of reversed (R) to number of normal (N) samples.

T/H range: range of unblocking temperatures/coercivities of components during thermal/AF demagnetization.

Comp.: component of magnetization.

N = number of samples or sites used in statistical analysis; N_0 = number of samples or sites exhibiting the component;

D = mean declination in degrees; I = mean inclination in degrees (directions uncorrected for tilt); k: Fisher precision parameter for mean direction; α_{95} = radius (degrees) of circle of 95% confidence about the mean direction; long. = East longitude of Virtual Geomagnetic Pole (VGP);

lat. = North latitude of VGP calculated from uncorrected mean directions; K = Fisher precision parameter for the pole. A_{95} = Radius of circle of 95% confidence about the VGP.

[a] Characters correspond to those in Fig. 1.

[b] Samples measured in China.

[c] Estimated by method in Cox (1970) and Coe et al. (1985).

TABLE 2

Paleomagnetic results for rock units sampled in Junggar, Xinjiang

Study site No.	Area (°N, °E)	Rock	Age	Strike/dip	Polarity ratio R/N	Temp. range (°C)	Comp.	Statistical analysis								
								N/N_0 coord.	directions				poles			
									D	I	k	α_{95}	long.	lat.	K	A_{95}
Kelamayi area																
109	45.6, 84.9	mud.+SS	J2	91/51	0/16	400–550	A	16/27 G	21.0	65.0	78.1	4.2	159.4	75.7	38.4	6.0
109B	45.6, 84.9	mud.+SS	J2	91/51 [a]	0/6	300–650	A	6/6 G	18.1	51.1	30.0	12.4	211.3	71.0	23.9	14.0
109+109B					0/22		A	22/27 G	20.0	61.3	43.3	4.8	175.8	75.7	28.2	5.9
								S	151.1	50.9	1.9	33.0	110.7	−9.4	2.0	30.2
110	45.5, 84.7	granite	C2–3	0/0	0/3	100–200	A	3/10 G	16.8	71.8	63.5	15.6	119.5	73.9	27.8	23.6
					10/0	550–650	B	9/10 G	213.4	−54.5	193.6	3.7	5.3	−62.7	106.9	5.0
111	45.0, 84.0	granite	C2–3	0/0	0/14	200–400	A	14/16 G	354.5	66.9	38.8	6.5	54.0	84.3	18.2	9.6
					16/0	500–600	B	16/16 G	214.6	−56.0	107.4	3.6	360.0	−62.8	8.08	4.1
Mean of component A (4 sites)							A	4/4 G	13.4	64.1	62.5	11.7	159.4	81.0	38.7	14.9
								S	39.7	71.7	1.4	156.7	143.0	65.6	1.3	180.
Mean of component B (2 sites)							B	2/2 G	214.0	−55.3	4831.9	–	3.2	−62.7	3206	–
Liusugou area																
112	45.6, 84.1	SS+cong.	P2	255/55	0/6	100–250	A	6/6 G	1.9	70.3	79.5	7.6	88.6	80.8	31.8	12.1
					4/0	570–650	C	4/6 G	224.0	−51.4	12.6	36.3	359.0	−56.4	8.9	44.1
113	45.6, 84.1	SS+cong.	P2	255/65	0/12	100–300	A	12/12 G	351.6	67.3	108.2	4.2	39.4	82.5	46.2	6.5
					7/0	560–650	C	7/12 G	216.4	−22.3	66.9	7.4	29.9	−44.1	73.6	7.1
114	45.7, 84.3	SS+cong.	P2	300/90	0/6	100–300	A	6/6 G	6.2	65.5	145.3	5.6	143.5	85.1	63.5	8.5
					3/0	400–600	C	3/6 G	241.2	−45.3	232.6	8.1	353.2	−38.6	226.4	8.2
Mean of component A							A	24/24 G	357.1	67.6	95.3	3.0	65.4	84.2	40.1	4.7
								S	355.6	−0.5	15.3	7.8	269.7	44.0	23.1	0.3
Mean of component C (averaging sites 113 and 114)							C	10/24 G	222.5	−29.6	22.8	10.3	18.4	−43.8	26.8	9.5
								S	221.6	20.3	31.8	8.7	40.6	−19.3	38.1	7.9
Mean of component C (3 sites mean)							C	3/3 G	226.2	−40.2	19.9	28.4	7.7	−46.6	26.3	24.5
								S	215.6	16.6	7.3	49.5	43.0	−26.3	12.3	36.7
Shanzhuquan area																
105	45.5, 84.7	tuff flows	C2	177/63	2/8	100–400	A	8/11 G	350.2	56.0	30.0	10.3	314.6	79.0	22.8	11.8
					11/0	450–575	B	11/11 G	219.7	−45.0	50.8	6.5	11.4	−53.3	41.2	7.2

106	45.5, 84.7	tuff flows	C2	50/32	4/3	200–450	A	9/14 G	17.0	53.2	10.2	16.9	208.8	73.5	7.6	20.0
					14/0	425–575	B	14/14 G	220.1	−53.9	70.4	4.8	0.2	−58.0	43.7	6.1
104	45.5, 84.7	tuff	C2	26/59	0/1	100–300	A	1/8 G	1.2	62.8	–	–	230.8	88.5	–	–
					6/0	450–560	B	6/8 G	229.0	−38.9	69.6	8.1	8.1	−44.2	97.1	6.8
102	45.5, 84.7	basalts	C1	26/82	0/4	100–300	A	4/6 G	358.6	69.6	203.3	6.5	77.6	81.8	95.0	9.5
					5/0	450–570	B	5/6 G	224.0	−49.8	276.3	4.6	2.6	−53.0	245.6	4.9
103	45.5, 84.7	basalts	C1	26/59	0/6	100–200	A	6/7 G	15.4	64.9	92.7	7.0	160.2	79.4	40.0	10.7
					7/0	525–570	B	7/7 G	234.7	−56.0	174.3	4.6	347.1	−48.8	112.9	5.7
101	45.5, 84.7	basalts	C1	26/82	0/7	200–400	A	7/9 G	357.3	60.3	26.9	11.8	88.3	20.9	26.9	11.8
					7/0	450–570	B	7/9 G	234.5	−59.8	56.8	8.1	341.2	−50.7	27.9	11.6
107A	45.5, 84.7	basalts	C2	206/45	6/0	400–600	B	6/6 G	249.2	−26.1	22.4	14.5	359.6	−24.7	27.6	13.0
107B	45.5, 84.7	basalts	C1	225/85	0/9	200–500	A	9/9 G	356.3	67.5	71.1	6.1	61.9	84.5	28.0	9.9
					9/0	400–600	B	9/9 G	238.5	−49.3	25.7	10.4	352.2	−43.9	23.2	10.9
107C	45.5, 84.7	basalts	C1	225/85	6/0	400–600	B	6/6 G	232.3	−40.2	21.8	14.7	4.8	−43.6	23.1	14.2
108	45.5, 84.7	basalts	C1	225/85	0/7	100–400	A	7/7 G	49.2	65.2	159.3	4.8	12.8	82.3	63.6	7.6
					7/0	425–600	B	7/7 G	210.6	−49.8	120.9	5.5	15.9	−62.1	87.6	6.5
Mean of component A (8 sites)							A	8/10 G	1.1	62.8	115.9	5.2	229.3	89.0	64.0	7.0
								S	42.1	37.2	1.8	60.0	190.6	46.2	1.8	60.0
Mean of component B (10 sites)							B	10/10 G	229.8	−47.4	42.0	7.5	0.1	−48.3	41.8	7.6
								S	223.1	−21.6	2.9	34.3	23.1	−42.1	3.1	33.2
Overall mean of component A from the three areas							A	15/16 G	3.1	64.8	124.4	3.4	133.0	87.3	66.1	4.7
								S	23.7	35.1	1.8	40.9	209.7	58.5	1.9	39.8
Overall mean of component B							B	12/12 G	229.0	−48.8	45.2	6.5	359.3	−49.6	43.4	6.7
								S	222.9	−29.2	3.3	28.5	19.0	−46.6	3.5	27.1
Synfolding +10 sites							B	11/11 SYN	227.7	−50.3	75.8	5.3	358.5	−51.0	58.3	6.0

Rock: rock type.

Polarity ratio: number of reversed (R) to number of normal (N) samples.

Temp. range: range of unblocking temperatures of components during thermal demagnetization.

comp.: component of magnetization.

N = number of samples or sites used in statistical analysis; N_0 = number of samples or sites exhibiting the component; coord.: G = geographic coordinates, S = stratigraphic coordinates; D = mean declination in degrees; I = mean inclination in degrees; k = Fisher precision parameter for mean direction; α_{95} = radius (degrees) of circle of 95% confidence about the mean direction; long. = East longitude of Virtual Geomagnetic Pole (VGP); lat. = North latitude of VGP. K = Fisher precision parameter for the pole; A_{95} = radius of circle of 95% confidence about the VGP.

[a] Overturned bed.

sites (Fig. 1b). Sample localities range in age from Mid-Carboniferous to Miocene. These localities exhibit well-exposed sections of limestone (mostly in quarries and roadcuts), sandstone, andesitic lave flows, tuffs, and basalts. Details of the paleomagnetic samples are summarized in Table 1.

Xinjiang

Sampling localities are confined to the West Junggar Basin near Kelamayi (= Karamay) city (Fig. 1c). With Siberia to the north, Kazakhstan to the west, and the Tarim block to the south, this block forms a triple junction within a convergent fold belt. West Junggar is also the most inboard among the blocks that make up the tectonic framework of China. The oldest rocks in this region are of Late Cambrian age (508 ± 60 Ma) and lie within the Tangbale ophiolite melange (Kwon et al., 1989). Middle Devonian strata consist of spilitic pillow lavas interlayered with radiolarian cherts and associated limestone horizons. Radiolaria found in the red cherts are *Tetrentactinia* cf. sp., *barysphaera Foreman, Cenosphaera* sp., *Heliodiscus* sp., and *Conadicus* sp. (Feng et al., 1989). Carboniferous rock units consist mainly of marine to continental volcanic rocks interbedded with clastics. No pillow-chert sequences have been found, suggesting that the west Junggar ocean basin began to close during the Carboniferous (Feng et al., 1989). Potassium-rich granitic plutons and their extrusive equivalents are common in this area, which have Permian K–Ar ages and Middle to Late Carboniferous U–Pb ages (Kwon et al., 1989). Permian rocks of west Junggar are mainly continental clastics and volcanic extrusives, and form an angular unconformity with the underlying Carboniferous strata. At many localities igneous dikes of Permian age intrude the Carboniferous formations. Triassic and Jurassic sequences are all continental clastics. In western Junggar, igneous rocks are widespread. The magmatic activity peaks during the Late Hercynian orogeny. Reported radiometric ages for igneous rocks of this orogeny range from 220 to 280 Ma (Chinese Academy of Geological Sciences, 1973; Zhao et al., 1983). Major regional unconformities in these strata occur between the late Carboniferous and Early Permian, the Late Permian and Early Triassic, and the Late Jurassic and Early Cretaceous. Major folding and regional deformation probably occurred during the Late Permian and Early Cretaceous.

We collected a total of 167 individual oriented samples from formations that range in age from Early Carboniferous to Middle Jurassic, including basaltic lava flows, densely welded tuff flows, granitic intrusions, and mudstone and sandstone beds (Table 2). We did not obtain any Triassic samples because we could not find any that were not severely weathered.

Sampling followed standard paleomagnetic practice, with in situ drilling by a portable gasoline-powered core drill and orientation using a magnetic compass mounted on an orienting device. We corrected for the effect of local magnetic anomalies on magnetic orientation by taking solar azimuths or by using azimuths of landmarks identifiable on maps.

Experimental techniques and data analysis

After specimen preparation, the natural remanent magnetization (NRM) of the specimens was measured. Usually companion specimens (A and B, respectively) cut from the same core were subjected to either stepwise thermal or alternating field (AF) demagnetization so that a comparison could be made between the coercivity and unblocking temperature spectra of the components of magnetization. A combination of AF and thermal treatments was required for some specimens. The hybrid AF–thermal demagnetization (i.e., thermal after AF demagnetization) was used in cases where the magnetization had been affected by isothermal remanent magnetization (e.g., lightning strikes). On the other hand, the hybrid thermal–AF demagnetization (i.e., AF after thermal demagnetization) was used for rocks with a hard viscous remanent magnetization that was not appreciably removed by AF treatment alone. It was also used for rocks where a thermo-chemical alteration caused problems at some critical temperature. Magnetic measurements were mainly performed with a 2G cryogenic magnetometer and a

Schonstedt spinner magnetometer in the paleomagnetic laboratory at the University of California at Santa Cruz, but a few samples from selected sites from Inner Mongolia were measured with a Molspin spinner magnetometer in Beijing, China. Demagnetization experiments were done using Schonstedt equipment. Magnetizations were determined by principal component analysis (Kirschvink, 1980) and the method of combined remagnetization planes and least-squares-best-fit-line data of McFadden and McElhinny (1988). Statistics on the distribution of paleomagnetic directions for each component at each site was calculated using standard Fisher (1953) statistics, and site mean directions of all demagnetized data were derived by giving unit weight to each mean sample direction.

Paleomagnetic results from Inner Mongolia

Our paleomagnetic sampling areas relevant to this paper are the two regions shown in Fig. 1b that are separated by the Xilamuren (= Xar Moron) River (Fig. 1b), and the directional data are summarized in Table 1. As mentioned in the introduction, the Xilamuren River is considered by many Chinese geologists to be a plate boundary or "suspect terrane" boundary because there are some remarkable differences in stratigraphy, petrology, structure, and paleontology between strata north and south of the river (Li, 1986; Wang and Liu, 1986). In order to test the geologic models (e.g., boundary of plates or suspect terranes), we chose sampling localities on both sides of this river. We will use northern and southern regions as geographic terms that encompass localities north and south of the Xilamuren River, respectively.

Results from the northern region

Cretaceous volcaniclastic rocks

We conducted a pilot study on Early Cretaceous volcaniclastic rocks in the vicinity of Dalai Lake (loality j in Fig. 1b). These rocks are part of an uppermost Jurassic to lower Cretaceous belt of volcanic rocks that characterizes this part of Inner Mongolia. In the study area, bedding attitudes were gently (less than 10°). In the laboratory, all samples revealed a reversed characteristic direction (Fig. 2a), usually after removal of a normal component at lower temperatures or alternating field treatments. The less stable normal component appeared to be a viscous remanent magneti-

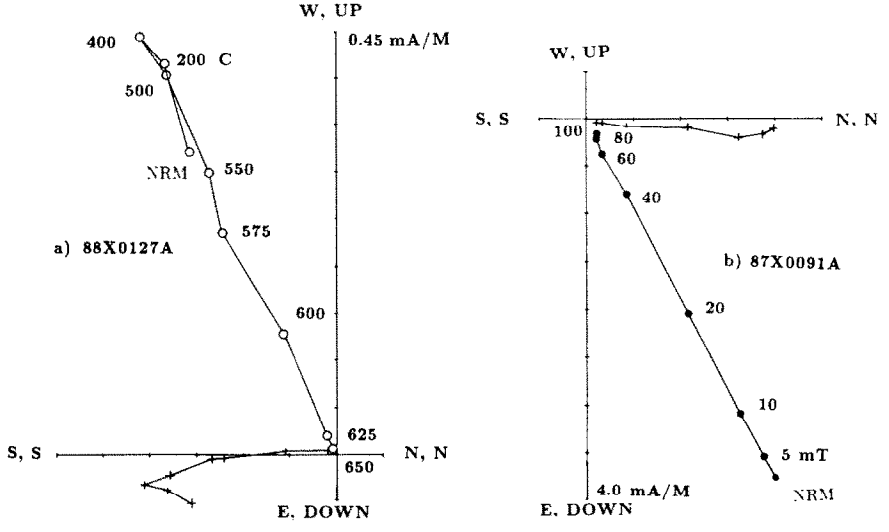


Fig. 2. Representative vector plots of Cretaceous rocks from Inner Mongolia: (a) thermal demagnetization on sample 88X0127A from the Dalai Lake area; (b) AF demagnetization on sample 87X0091A from the Pingzhuang area. Directions are plotted in geographic (= in situ) coordinates. Crosses represent vector endpoints projected onto horizontal plane; solid circles represent endpoints projected onto the north–south vertical plane.

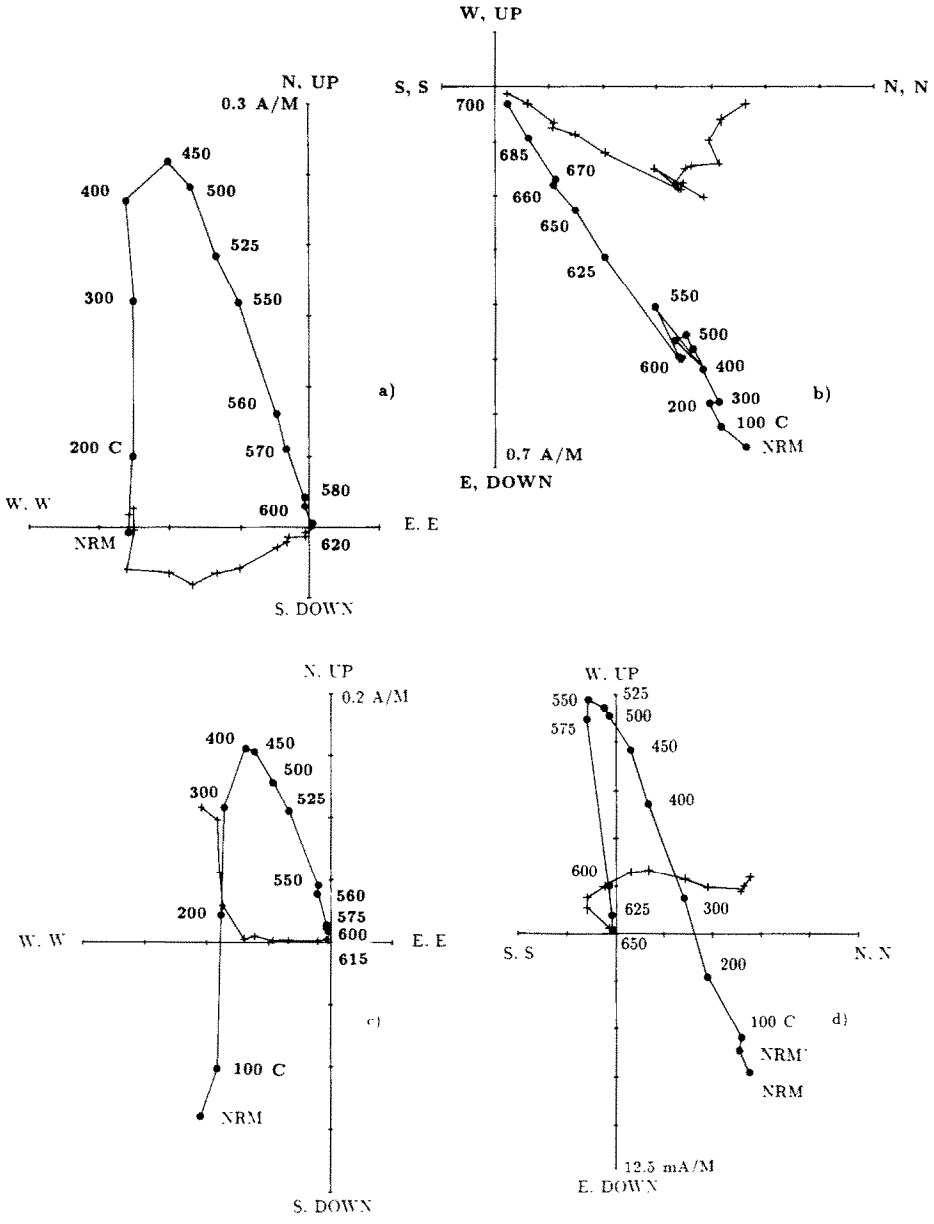


Fig. 3. Representative thermal demagnetization vector plots of Jurassic rocks from Inner Mongolia: (a) Manzhouli ara; (b) Hua-an area; (c) Zalantun area; (d) Balinzuo Qi area. NRM′ indicates that NRM was measured after sample had been stored in zero field for two weeks. Directions are plotted in geographic coordinates, showing normal overprints and reversed and normal characteristic components. Crosses and solid circles same as Fig. 2.

zation (VRM) of the Recent geomagnetic field in some cases and, as shown by large scatter in the directions and exceptionally strong intensities, lightning-induced isothermal remanent magnetization (IRM) in others. Often the intensity of magnetization increases by a small percentage as normal VRM is removed. Thermal demagnetization and IRM experiments showed that the magnetic carriers of these samples are both magnetite and hematite. Thus, even though no fold and reversal tests are available, the characteristic direction is probably primary.

Late Jurassic andesite and tuff flows and tuffaceous sandstone

Late Jurassic volcanic rocks are well developed

TABLE 3

Characteristic site-mean directions and poles for Inner Mongolia

	P	n/N_0	Coord.	Direction				Pole position			
				D	I	α_{95}	k	long.	lat.	A_{95}	K
Tertiary sites (A1–6, L1–7 in Table 1)											
Normal	N	8/13	G	355.4	52.8	13.2	18.6	321.2	81.8	16.8	11.8
			S	359.6	53.7	6.9	65.9	298.0	83.1	9.3	36.1
Reversed	R	5/14	G	161.8	−50.5	33.8	6.1	204.6	−76.2	30.4	7.3
			S	175.8	−67.4	9.4	67.1	279.7	−78.3	12.7	37.3
Mixed	M	13/13	G	350.3	52.2	12.9	11.2	352.9	81.2	13.8	9.9
			S	358.5	59.0	6.1	47.0	29.2	88.6	8.2	26.3
Cretaceous sites (A7–10, J1–2)											
Normal	N	4/6	G	13.3	71.2	8.9	107.0	146.5	73.1	14.6	40.5
			S	4.4	57.1	6.6	193.2	258.4	84.8	8.3	123.4
Reversed	R	2/6	G	169.3	−63.0	–	324.2	183.0	−81.8	–	157.0
			S	184.3	−61.9	–	1125.5	85.6	−84.0	–	465.9
Mixed	M	6/6	G	7.0	68.7	6.7	100.6	147.9	80.2	12.5	29.8
			S	6.8	58.5	4.7	207.7	249.5	82.9	5.7	138.2
Jurassic sites											
(I1–14)											
Normal	N	4/14	G	13.3	57.4	8.0	132.3	252.1	75.3	10.2	82.0
			S	25.0	54.8	8.1	128.5	235.7	67.4	8.7	113.8
Reversed	R	10/14	G	213.6	−40.3	4.6	113.4	60.1	−52.8	4.3	128.7
			S	216.3	−54.7	4.7	108.2	41.2	−60.2	5.8	70.2
Mixed	M	14/14	G	29.2	45.5	6.1	43.6	241.8	59.3	6.4	39.0
			S	33.1	54.9	3.9	104.0	224.6	62.4	4.9	66.8
(F1–3, F6–9, G1)											
Normal	N	7/8	G	31.6	70.8	9.9	38.1	178.9	68.8	17.0	13.5
			S	15.4	51.4	6.8	90.9	260.0	71.6	7.8	60.2
Reversed	R	1/8	G	262.1	−66.1			358.7	−38.2	8.0	28.6
			S	200.9	−66.8			17.0	−76.4	8.1	57.1
Mixed	M	8/8	G	38.7	71.1	9.6	34.5	179.0	64.9	16.4	12.4
			S	15.9	53.3	6.9	65.3	254.8	73.0	7.8	51.1
Site D1	R		G + S	212.3	−51.8	2.9	702.8	40.5	−62.6	3.0	655.7
Site E12	R		G	246.0	−79.0	7.8	60.2	328.9	−48.5	14.6	18.1
			S	197.4	−54.2	7.8	60.1	58.0	−73.8	9.8	39.2
Overall average (4 areas, see text and Table 1)											
	M	4/4	G					247.5	58.4	110.6	1.7
			S					231.6	68.5	9.5	93.6
Late Permian sites (B1–2, B4–7, D2–6)											
Normal	N	4/11	G	347.1	44.2	14.0	43.9	334.8	70.8	11.2	68.6
			S	330.1	26.4	13.0	48.6	350.0	51.2	10.9	71.9
Reversed	R	7/11	G	161.1	−32.9	11.3	29.5	157.3	−60.9	11.5	28.3
			S	157.4	−19.5	11.1	30.6	156.8	−52.0	9.6	40.4
Mixed	M	11/11	G	343.1	37.1	8.4	30.5	336.6	64.5	7.9	34.0
			S	334.8	22.0	7.9	34.2	341.7	51.9	6.8	46.3
Late Carboniferous sites (E9–11)											
Reversed	R	29/29	G	150.8	−32.3	15.7	3.9	176.2	−59.7	11.1	6.8
			S	148.6	11.5	6.2	19.5	158.1	−32.6	4.4	38.7

P (polarity): N = normal, R = reversed, M = both normal and reversed; n/N_0 = number of samples, sites, areas, or regions used in statistical analysis/total of the samples, sites, areas, or regions exhibiting the characteristic component; Coord.: coordinates—see Table 1 for explanation of other abbreviations.

in the northern region. Radiometric ages reported for these volcanic rocks range from 139 to 199 Ma, with a favored figure close to 155 Ma (Chinese Academy of Geological Sciences, 1973; T.D. Wang, 1986). Our Late Jurassic rock collection in the northern region covered a wide area. Going from north to south, sampling localities are distributed from Manzhouli (locality i in Fig. 1b) through Zalantum (locality g) and Hua-an (locality f) to Balinzuo Qi (locality e), a region spanning 5–6° of latitude and longitude. Rocks included andesitic lava flows, tuffs, and tuffaceous sandstones.

Results from Manzhouli area. A total of 14 tuffaceous sandstone units that span about 6000 m of section were sampled at two localities. In the northern locality, which is near the border with the Soviet Union, a large scatter in the direction of natural remanent magnetization was observed from most of the samples. Demagnetization in alternating fields ranging from 20 to 50 mT or at temperatures up to 500°C effectively removed this random overprint and revealed a characteristic direction. The characteristic direction is directed southwesterly with negative inclination and behaves well magnetically (Fig. 3a). At the southern locality, which is about 20 km from the northern sampling section, the stable component is normal in every sample and is obtained predominantly in the 600–675°C interval, showing that hematite is the main carrier of the characteristic component. As can be seen in Table 3, the fold test is positive at more than 95% confidence, indicating that remanent magnetization was acquired before folding in the Late Cretaceous. Moreover, the mean normal and reversed directions are not significantly different from antipodal, affording a positive reversal test.

Results from Zalantun and Hua-an areas. About 600 km to the southeast of the Manzhouli area, we sampled another thick section of Late Jurassic andesites and tuffites that is well exposed in two separates localities. The first locality is in Saligou, about 10 km east of downtown Zalantun. The second locality is located in Hua-an, about 100 km southeast of Zalantun. The paleomagnetism of these samples is very similar to those of their counterparts in the Manzhouli area. Specifically, the random overprint component, which is probably induced by lightning, is usually present. Both AF and thermal demagnetization methods remove this random overprint component and isolate the characteristic component. The characteristic directions isolated from 7 sites (F1–3, F6–9, Table 3) at the Hua-an locality are all normal (Fig. 3b), but at the Zalantun locality (site G1) the direction is reversed (Fig. 3c). Although only one site exhibits a reversed direction, the data probably give a reasonable indication of the time-averaged direction. The generalized fold test is positive (Table 3), further suggesting the characteristic component is probably primary.

Results from Balinzuo Qi area. Some 30 km southeast of downtown Linxi in the Balinzuo Qi area, we drilled nine tuff samples from roadcut exposures along the Linxi–Xingfuzilu road. The flows dip 30° to the north and cores were very difficult to drill. The lack of continuous outcrop made it difficult to determine how many flows the nine cores sampled. In the laboratory, thermal demagnetization to 500°C successfully cleaned a secondary component with a direction very close to that of the present dipole field, indicating a VRM origin. Above 500°C, the demagnetization curves all reached stable end points (Fig. 3d), with reversed, southerly declinations and negative inclinations that agree well with the directions of coeval rocks from the more northerly localities discussed above. Although the number of flows could be very few and there is no other independent evidence to show that a good time average of the ancient field has been obtained by these samples, we have no reason not to believe that the characteristic direction is primary.

Late Permian sediments

In the vicinity of Xingfuzilu of Balinzuo Qi (locality e in Fig. 1b), Late Permian sediments were targeted in order to compare results with Late Permian directions from the southern region. Four measured sites were from a section of metasediments, mainly sandstones. Progressive thermal demagnetization on rocks from two sites

(E5 and E13 in Table 1) isolated only a single stable component of normal magnetization. Rocks from the other two sites (E3 and E4), on the other hand, exhibited two components of magnetization. The low unblocking temperature component is northerly and downward, which is generally close to the normal recent field direction in geographic coordinates. The higher unblocking temperature component is southerly and upward. No fold test is applicable to these sandstone sites so that the timing of this reversed magnetization is conjectural. However, the fact that the pole position corresponding to the data in stratigraphic coordinates falls close to that of the Tertiary, Cretaceous, and normal overprint directions seen elsewhere in Inner Mongolia, suggests that the Late Permian data from the northern region may be an overprint acquired in Mesozoic to Cenozoic time.

Early Permian rocks

Two Early Permian sandstone sites were sampled at Hua-an (locality f in Fig. 1b), near the border with Helongjiang province. Thermal demagnetization of rocks from the two sandstone sites only revealed a single component (north and down). We also sampled six sites from fresh exposures of Early Permian andesites at two localities near Balinzuo Qi (locality e in Fig. 1b). At samples from site E1, two components of magnetization are present. The first is the present-day field component. The second component, however, is skewed somewhat to the east and also has positive inclination. Samples from other sites exhibited rather different demagnetization behavior. Most samples revealed only one component, shown by stable end points on vector plots, but the directions are very scattered. In a few cases, the stable direction is revealed after removal of a seemingly reversed component. In view of the complicated magnetic behavior of these rocks and the large scatter in directions, it appears that these rocks may have suffered different degrees of overprinting that we have not been able to adequately isolate from an earlier remanence.

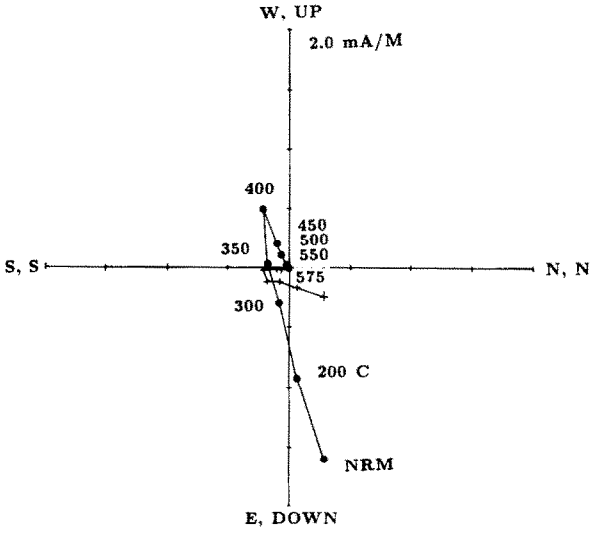


Fig. 4. Typical vector plot for Carboniferous limestone from the Tianshan area of Inner Mongolia. A normal overprint and a reversed characteristic component exist in this sample. Directions are plotted in geographic coordinates. Crosses and solid circles same as Fig. 2.

Carboniferous limestone

Across the Xilamuren River in the quarry of the Tianshan concrete factory (locality e in Fig. 1b and E8–12 in Table 1), we sampled four sites from a Carboniferous limestones and dolomite section. Although the initial NRM intensity was very weak, on the order of 2×10^{-4} A m^{-1}, the paleomagnetic signals from these limestones are quite interesting. Upon progressive thermal and AF demagnetization, a low unblocking temperature component was removed, revealing a characteristic component in three sites (Table 1 and Fig. 4). In site E8, however, only the low unblocking temperature component can be resolved. The directions of the characteristic component cluster around a mean with southeasterly declination and shallow downward inclination (Table 3). The fold test is positive, as shown by a factor of 5 increase in the precision parameter k after tectonic correction, which suggests an ancient origin for this component (Table 3).

About 40 km northeast of Hailaer city, a Middle Carboniferous limestone section is well exposed in a quarry near the Ridangshan area (locality h in Fig. 1b). The characteristic component with southerly declinations and upward inclinations is revealed after the removal of a normal Recent field component. The mean directions of

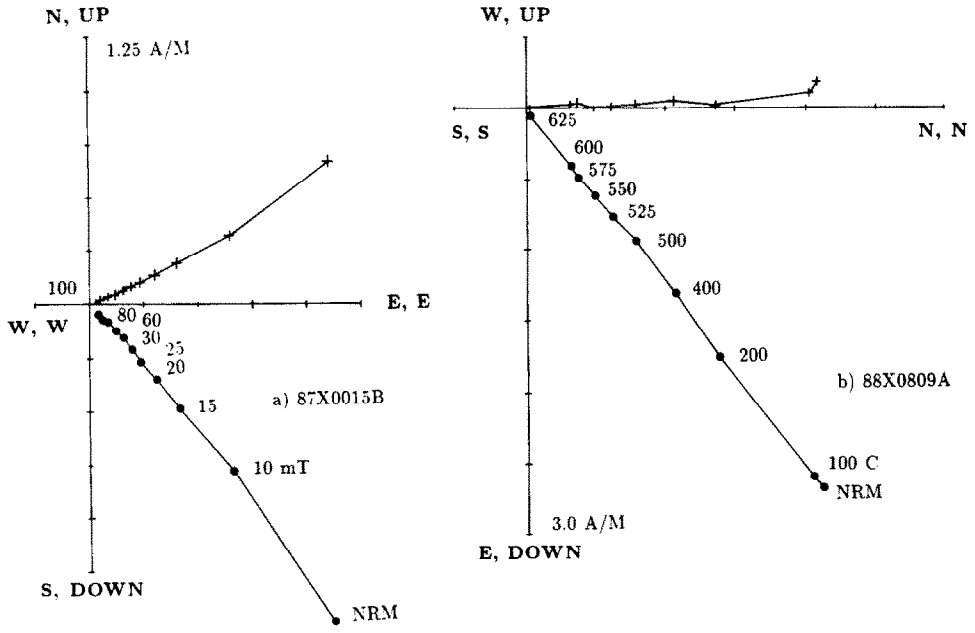


Fig. 5. Representative vector plots of Tertiary basalts: (a) AF demagnetization on sample 87X0015B from the Pingzhuang area; (b) thermal demagnetization on sample 88X0809A from the Jining area. Directions are plotted in geographic coordinates. Crosses and solid circles same as Fig. 2.

both components cluster very well (Table 1), but no fold test is available.

Some 300 km to the southeast in the vicinity of Zalantun city (locality g in Fig. 1b), we visited a Late Carboniferous sandstone formation. Only five samples were taken because of the apparent surface weathering. Like its counterpart in Hailaer, the lower unblocking temperature component resembles a Recent field overprint. The higher unblocking temperature has southerly declinations and upward inclinations, but was only resolved at demagnetization temperatures less than 400 °C, considerably less than that for other rock units.

Results from the southern region

Almost all samples collected from this region revealed a present-day to Recent field magnetic component, and in some samples this was the only component that could be determined. However, many samples showed a second component that has been summarized in Table 1 (labeled as component 2) and will be discussed below.

Tertiary basalts

Two localities in the Pingzhuang area (locality a in Fig. 1b) of eastern Inner Mongolia, and one locality of Hannuoba Basalt Formation, in the Jining area (locality 1 in Fig. 1b) of central Inner Mongolia were sampled. K–Ar ages of these basalts range from 14.1 to 25.4 Ma (Deng et al., 1988; Liu, 1988). Both AF and thermal demagnetization methods were equally successful in the great majority of cases, except for a small fraction of samples in which thermal demagnetization was necessary to remove a secondary overprint carried by hematite. All samples yielded a well defined characteristic magnetization, as shown by demagnetization plots that reached stable endpoints (Fig. 5). The paleomagnetic results are given in Table 1 (sites A1–6 and L1–7).

Cretaceous tuffaceous sediments

We sampled four Early Cretaceous tuffaceous sediment sites (A7–10 in Table 1) from an open-pit coal mine near Pingzhuang (locality a in Fig. 1b). At site A10, samples were magnetically straightforward. A characteristic component, defined by univectorial decay toward the origin, was isolated using both AF and thermal demagnetization (Fig. 2b). At the other three sites, however, the samples displayed complicated magnetic behavior. After removal of a soft viscous overprint, a more stable

component is sometimes isolated by 500 °C thermal demagnetization that is very similar to the characteristic component from site A10. Above 500 °C, however, almost all samples altered drastically, as shown by erratic directional changes and orders of magnitude increase in the intensity of remanence. We also observed abrupt changes in magnetic susceptibility after successive heatings to 600 °C. We do not fully understand this effect yet. Further paleomagnetic and rock magnetic studies, as well as microscopic examination and magnetic separation, are currently in progress to understand these samples better.

Late Jurassic granite

We sampled a granitic body near Shizhuangfang (locality d in Fig. 1b). Radiometric dates are unavailable for this granite, but its age based on geologic relations is inferred to be Middle to Late Jurassic, according to our field guide from a local geologic team. Thermal demagnetization on most samples effectively removed a random overprint and revealed a stable characteristic magnetization that is southerly and up. Judging from the unblocking temperatures, the coercivities, and the IRM experiments, the stable component is probably carried by magnetite.

Late Permian sediments and volcanic rocks

We sampled Late Permian strata from two localities in Wengiute Qi. In the Wudan area (localities b in Fig. 1b), a well bedded section of redbeds and thick andesitic lava flows crops out along the banks of the north Ranfangdi ravine (about 6 km north of Wudan). About 80 km north, fresh exposures of andesitic lava flows are well exposed along roadcuts near Shizhangfang (locality d). Rocks from these two areas exhibited similar magnetic behavior during progressive thermal demagnetization, revealing a characteristic component with southeasterly or northwesterly declinations and shallow upward or downward inclinations, respectively (Fig. 6). *F*-ratio criteria indicate that these two directions are not different at the 95% confidence level if one polarity is inverted to the other. At two sites from the Wudan area (sites B2 and B7), about 60% of the samples revealed the characteristic component in vector plots, whereas the remaining samples showed evidence for this component in their demagnetization planes. The means for the combined data set of lines and planes, obtained by using the method of McFadden and McElhinny (1988), are summarized in Table 1.

Early Permian sandstone and mudstone

Some 200 km southeast of Erlianhaote, we sampled four sites from a sandstone and mudstone formation that is ascribed to the Early Permian (locality k in Fig. 1b). Rocks from all four sites yielded a principal component with northerly declination and moderately downward inclination. A few samples revealed a higher unblocking temperature component, but the direction is rather poorly established. A fold test on the principal component is entirely negative, indicating that it is a secondary remanence imposed after folding, probably a Recent field overprint.

Carboniferous limestones and redbeds

We sampled four Carboniferous sites from Aoji (locality c in Fig. 1b), two in redbeds and two in a sandstone–shale sequence derived from volcaniclastic material. After removal of a normal overprint by 400 °C thermal demagnetization, a higher unblocking temperature component was isolated with northeasterly or southwesterly declinations and moderate downward or upward inclinations in geographic coordinates, respectively. Unfortunately, essentially no fold test is applicable to these sites, and therefore the age of the remanence is uncertain.

Another sampling locality is located in Minganaobao of central Inner Mongolia (locality k in Fig. 1b). Four limestone and one sandstone sites were sampled. In general, the outcrops were poor and errors could be large both in age of the formation and the bedding corrections. These rocks behaved magnetically less satisfactorily than other coeval rocks to the east. Nevertheless, two components of magnetization were evident in these rocks. The first component has a low unblocking temperature with directions close to the present field direction. The second component is revealed above 400 °C, with southwesterly declination and shallow upward inclination. The first component

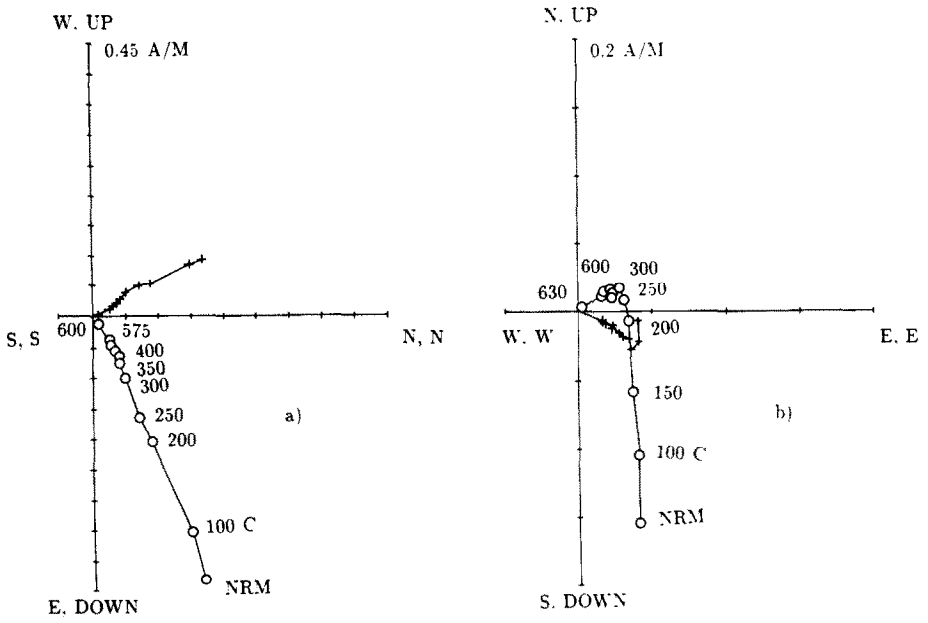


Fig. 6. Representative thermal demagnetization vector plots of Late Permian rocks from the southern region: (a) normal characteristic component; (b) normal overprint and reversed characteristic component. Directions are plotted in geographic coordinates. Crosses and solid circles same as Fig. 2.

clearly failed the tilt test, but the second component passed marginally.

Discussion of paleomagnetic results from Inner Mongolia

Both overprinted and primary components are revealed in the results from Inner Mongolia. For Tertiary, Cretaceous, and Jurassic rocks, the overprints appear mainly to be Recent field IRM and VRM. For other rock units, overprints may be CRM in some cases and TRM in others, as indicated by the unblocking temperature range and by polarity distribution patterns. The overprint component (labeled as component 1 in Table 1) in all but two cases clearly fails the fold test, showing a post-folding age (Table 4).

The Early Permian paleomagnetic data were derived from widely distributed volcanic and sedimentary units along the banks of the Xilamuren River. Demagnetization of these rocks revealed a complicated behavior of magnetization. Some rocks displayed a multicomponent nature, with a characteristic component (component 2 in Table 1) isolated after removal of a soft component. However, because the internal inconsistency is very large and no fold test or other independent evidence i available to demonstrate the age of magnetization, we exclude the Early Permian data from further discussion.

Late Permian and Carboniferous rocks from sampling localities north of the Xilamuren River revealed a multicomponent nature, but no local fold test was applicable to the magnetization. Although the regional tilt test is inconclusive, the directions of the higher unblocking temperature component (component 2) in geographic coordinates are close to those of the Late Jurassic rock units from this region, leaving the possibility open that these Paleozoic rocks may have been overprinted in Late Jurassic time. Consistent with this hypothesis is the fact that this component is not present in Cretaceous or younger rocks. It is interesting to note that the intensity of this possible Late Jurassic remagnetization increases toward the north, which coincides with the widespread occurrence of andesitic rocks in this region and may be associated with suturing processes between the Chinese continental blocks and Siberia during the Mesozoic. Further discussion about this point will be given in a later section.

The characteristic directions isolated from the

Carboniferous limestone section in the Tianshan concrete factory, which is located north of the Xilamuren River but more eastward than the localities discussed above, also have a uniform reversed polarity. The characteristic direction was revealed after removal of a secondary overprint and passes the fold test. Because the earliest folding that affected these limestones is of Late Permian age, the remanence could represent the primary magnetization in a field of the Kiaman Long Reversal Superchron in the Late Carboniferous. We tentatively interpret the corresponding

TABLE 4

Overprinted components in Inner Mongolia

Area	Age	N	Coord.	D	I	α_{95}	k	Long.	Lat.	A_{95}	K
Component 1 (see Table 1)											
b	P2	5	G	4.8	64.8	4.9	249.1	156.7	84.4	6.4	145.7
			S	21.5	58.5	40.9	4.5	215.5	73.5	57.6	2.7
c	P1–C	7	G	23.7	65.5	8.6	51.9	181.5	72.2	12.6	24.1
			S	15.0	36.7	49.7	2.4	253.8	68.6	56.3	2.1
d	J3–P2	6	G	3.4	63.7	4.5	222.4	159.9	86.5	6.1	121.6
			S	329.7	50.8	12.1	31.9	14.0	64.8	15.2	20.3
e	J3–C	13	G	353.8	68.1	6.9	37.1	96.9	80.6	9.8	18.9
			S	246.8	65.6	32.9	2.6	82.4	18.9	40.1	2.0
f	P1	2	G	8.3	72.8	–	1035.2	144.0	77.6	–	390.3
			S	239.3	80.0	–	187.1	102.9	35.1	–	54.3
g	J3–C	4	G	358.5	68.6	6.8	183.6	112.0	85.8	9.3	98.9
			S	21.5	70.4	90.7	2.0	182.5	77.1	180.5	1.3
h	C2	3	G	7.8	63.5	–	168.4	243.7	83.7	–	84.2
			S	51.7	59.3	–	164.1	203.4	52.9	–	107.3
i	J3	3	G	10.0	65.8	–	20.4	206.5	82.9	–	8.9
			S	19.9	69.4	–	27.6	184.1	77.8	–	10.6
j	K1	2	G	12.2	69.6	–	278.0	172.1	81.0	–	101.1
			S	28.7	65.4	–	135.1	200.0	70.8	–	52.7
k	P1–C2	8	G	353.2	61.8	3.3	283.2	26.7	84.9	4.3	165.8
			S	89.0	50.0	18.3	10.1	176.3	22.1	19.8	8.8
Overall		10	G	5.4	66.7	3.0	266.6	162.8	84.7	4.5	115.1
			S	19.7	70.2	16.4	9.7	157.2	73.5	25.4	4.6
Possible Late Jurassic overprint (Component 2, Table 1)											
h	C2	4	G	179.1	−55.9	6.9	175.8	124.0	−77.4	8.6	114.9
			S	214.0	−57.0	7.5	151.7	42.1	−63.2	9.2	99.9
k	C2	5	G	224.9	−41.2	13.6	32.7	36.4	−49.0	15.9	24.0
			S	262.2	−28.5	11.5	45.6	16.7	−16.0	9.3	68.9
Mean	C2	9	G	207.9	−50.0	13.3	16.0	51.6	−66.2	16.8	10.3
			S	246.5	−43.5	16.5	10.7	23.5	−37.4	18.6	8.6
e	P2	2	G	244.5	−26.1	–	191.1	36.8	−27.8	–	275.0
			S	184.4	−55.5	–	134.7	93.0	−81.5	–	74.6
g	J3?	1	G	248.3	−38.7			33.3	−31.2		
			S	338.3	−46.7			321.4	11.4		
Mean		12	G	219.0	−46.4	12.8	12.5	45.1	−57.2	15.3	9.0
			S	240.2	−43.9	17.4	7.2	28.3	−43.2	18.8	6.3

Area: characters correspond to those in Table 1.

Ages: ages of formations sampled that exhibited overprint component.

N = sites, area, or regions that exhibited overprinting; Coord.: coordinates—see Table 1 for explanation of other abbreviations.

pole as a Late Carboniferous pole for Inner Mongolia. However, because of the limited number of samples, we do not regard this pole position as firmly established. More detailed investigation of equivalent formations north and south of the Xilamuren River is needed to obtain confirming results.

The characteristic components of Tertiary, Cretaceous, and Late Jurassic rocks pass both local and regional fold tests at more than 95% confidence. Thus, the remanent magnetization was acquired before folding. Moreover, closely antiparallel polarities were found in all cases, indicating that enough time had been spanned by these rocks to obtain the ancient dipole field direction. The characteristic magnetization is most volcanic rocks is carried by both magnetite and hematite, indicating the hematite was probably formed by auto-oxidation processes during primary cooling of the volcanic rocks. The successful and reversal tests, and the fact that the characteristic component is carried by magnetic minerals that are capable of preserving an early remanent magnetization over a long period of time, strongly suggest that the magnetization directions derived from these rocks are primary and suitable for calculating the corresponding paleomagnetic poles for the Tertiary, Cretaceous, and Late Jurassic. The consistency of the results implies that these widely separated sampling areas in Inner Mongolia were a coherent tectonic unit since at least Late Jurassic time.

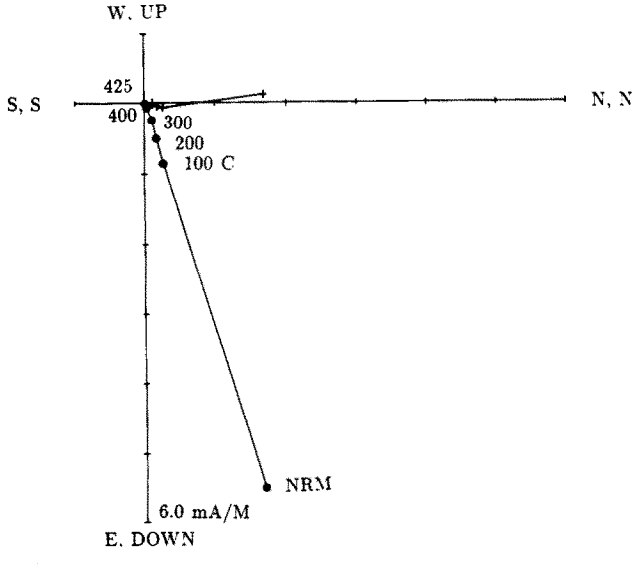


Fig. 7. Typical vector plot showing goethite behavior in Middle Jurassic rocks from Junggar. Directions are plotted in geographic coordinates. Crosses and solid circles same as Fig. 2.

The Late Permian results from localities south of the Xilamuren River also yielded positive fold and reversal tests results (Table 3), and are in good agreement with that of the NCB (Table 5). This indicates that the region south of the Xilamuren River was a part of the NCB in Late Permian time. Our paleomagnetic data are still too few to allow a complete reconstruction of the apparent polar wander part for Inner Mongolia; however, when compared with available paleomagnetic data and geologic, geochronologic or paleontologic observations from adjacent blocks, they are sufficient to put important constraints on the tectonic evolution of Asia, as will be discussed in a later section.

Paleomagnetic results from the Junggar basin, Xinjiang

Mid-Jurassic sandstone and mudstone

Samples were collected from Mid-Jurassic sandstone and mudstone sections located northeast of Kelamayi city (sampling location: 84.9°E, 45.6°N, see Fig. 1c). An angular uncomformity between Late Jurassic and Early Cretaceous formations is clearly visible at the outcrop, showing that folding and tilting occurred before Early Cretaceous. A total of 27 samples were subjected to progressive thermal and alternating field demagnetization. AF demagnetization had little or no cleaning effect. Thermal demagnetization at temperatures up to 200°C resulted in a pronounced drop in most sandstone samples (Fig. 7). These characteristics point to the existence of some goethite. The direction of the remanence unblocked between 20 and 200°C ($D = 7.2°$, $I = 67.3°$) is close to both the geocentric dipole field and the present day field direction for the sampling locality, suggesting that this goethite-derived component may be a recent CRM overprint. Unlike the sandstone, mudstone samples did not indicate the presence of goethite. In both sandstone and mudstone samples, a characteristic compo-

nent is revealed between 300° and 650° C thermal demagnetization. The characteristic component has a northerly and moderate-to-steep-downward direction in geographic coordinates. The fold test is clearly negative (Table 2), showing that the characteristic component is an overprint magnetization acquired after folding in the Late Jurassic. For convenience, this component is hereafter termed the "A" component.

Late Permian sandstone and conglomerate

Three Late Permian sandstone and conglomerate sites were collected near Liusugou (84.16–84.33° E, 45.58–45.65° N, Fig. 1c), about 150 km southwest of Kelamayi. These sites occur in the Late Permian Zhidi formation that contains the fossils *Cordaites* sp. *Zalessky, Sphenopteis* sp., *Pecopteris anthriscifolia (Goepp.)*, and *Noeggerathiopisis derzavinii Neub.* (Xinjiang Stratigraphy Tables Compilation Team, 1981; Wu, 1983). The sandstone is well bedded and reddish in color. In general, rocks from these three sites exhibited very similar demagnetization properties in the laboratory. The most striking feature is that a multicomponent magnetization exists in all sandstones, but only in some of the conglomerates. As shown in Fig. 8, the first component dominates the magnetization of the sandstone and has a low unblocking temperature (~ 200° C). The directions of this component cluster well and are generally close to that of the A component from the Jurassic rocks. The second component has a higher unblocking temperature (> 500° C), and is isolated by further progressive thermal demagnetization only. The higher unblocking temperature component has a southerly declination and moderate upward inclination in geographic coordinates (labeled as component C in Table 2). Although the scatter of the mean direction of this C component is quite large, in all cases it is of reversed polarity.

The common component in the conglomerates is directly north with downward inclination, which again is very close to the A component seen in other strata. No sample showed the high unblocking temperature component direction (south up). Because the low unblocking temperature component is present in both sandstone and conglomerate, but the higher unblocking temperature component can be seen only in the sandstone, the former is definitely secondary and the latter might be primary.

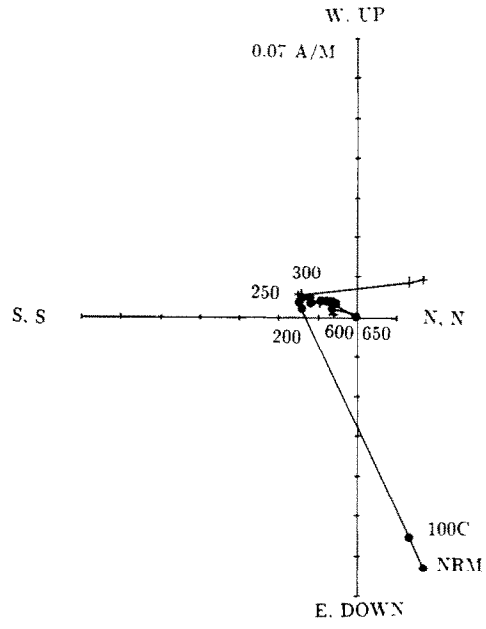


Fig. 8. Demagnetization behavior of Permian sandstone sample from Liushugou. Directions are plotted in geographic coordinates. Crosses and solid circles same as Fig. 2.

Middle to Late Carboniferous granitic bodies

As a check for potential thermal remagnetization due to the effects of igneous intrusions in the nearby strata, we sampled two spectacular A-type granitic bodies (Coleman, 1989) that were distributed north and northwest of Kelamayi. These pink and grey granitic bodies are rich in biotite and feldspar. Reported K–Ar ages range from 200 to 284 Ma, with a favored figure toward the older ages (Feng et al., 1989). However, more recent Pb-isotope studies show the more correct age to be about 320 Ma (Kwon et al., 1989), which points to the Middle Carboniferous. The Pb age is highly preferred because it is in good agreement with the observed geologic relations in the field; these granites intrude Carboniferous strata, and are covered by strata of Early Permian age.

The demagnetization behavior of these granite samples is rather straightforward. Both progres-

sive thermal and AF demagnetization isolate two distinct components of magnetization. The direction of the first component in geographic coordinates is statistically indistinguishable from that of the A component in the Jurassic sediments, suggesting that it may have been overprinted in the same paleofield and perhaps in the same manner. The second component is resolved at temperatures above 400°C and has southerly declination and upward inclination. This component, defined by the univectorial decay of the demagnetization path toward the origin of a vector diagram, is very stable and well clustered. It is distinct from the higher unblocking temperature component seen in the Late Permian sandstone discussed above, and therefore is a new feature in our paleomagnetic data from Junggar. For simplicity, we will call this component the "B" component.

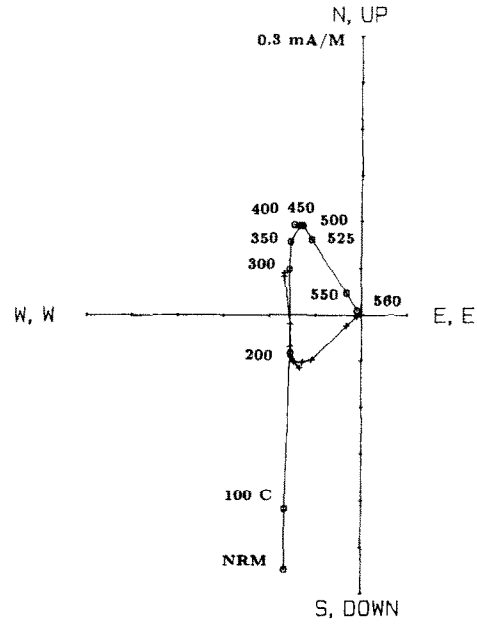


Fig. 9. Representative vector plot of Carboniferous tuff sample from the Shanzhuquan area. Directions are plotted in geographic coordinates. Crosses and solid circles same as Fig. 2.

Early and Middle Carboniferous tuff and basaltic flows

We sampled a thick Carboniferous section near Shanzhuquan, about 50 km southwest of Kelamayi. This section is a key sampling locality for three reasons. First, strata in this section spanned much of the Carboniferous, consisting of continuous formations of densely welded tuff and basaltic flows. Second, strata are folded and the bedding attitudes provide favorable fold geometries to allow a fold test on the remanent magnetization. Third, this area was re-mapped in detail by geologists from the Institute of Geology of the Academia Sinica during this investigation, so information concerning the structure and stratigraphy was readily available. Another striking feature of this sampling locality includes several Permian dikes that locally cut the Carboniferous volcanic sequence and quartz veins that are commonly present in the rocks, all indicating that the strata may have undergone late thermal events.

We collected a total of six sites of Early Carboniferous basalt flows at discrete locations and four sites of Middle Carboniferous tuff in the Shanzhuquan section. Although these Carboniferous rocks have quite different structure, lithology, and ages, they are strikingly similar to each other in demagnetization behavior. Specifically, both A and B components are present in these samples (Fig. 9). The B component, with monotonously reversed polarity (southwesterly declination and upward inclination), was isolated after removal of the A component during both thermal and AF demagnetization. Magnetite appears to be the dominant magnetic carrier in these volcanic rocks, as suggested by results from thermal and AF demagnetization, IRM acquisition, and thermal demagnetization of saturation IRM.

The mean direction of the B component, before tectonic correction, is declination $D = 229.8°$, inclination $I = -47.4°$, and 42.0. After bedding corrections are made, the new mean direction is $D = 223.1°$, $I = -21.6°$, and $k = 2.9$. The decrease in k indicates an increase in scatter and thus a failure of the standard fold test (Graham, 1949). Under the assumption that the B component isolated from these rocks was overprinted by the same mechanism in the same paleofield, we applied the incremental fold test on different sites. The results clearly show that the B component in the Shanzhuquan area was acquired toward the end or after the regional deformation. One of the strongest synfolding examples is found in the locality where sites 106 and 107A were sampled.

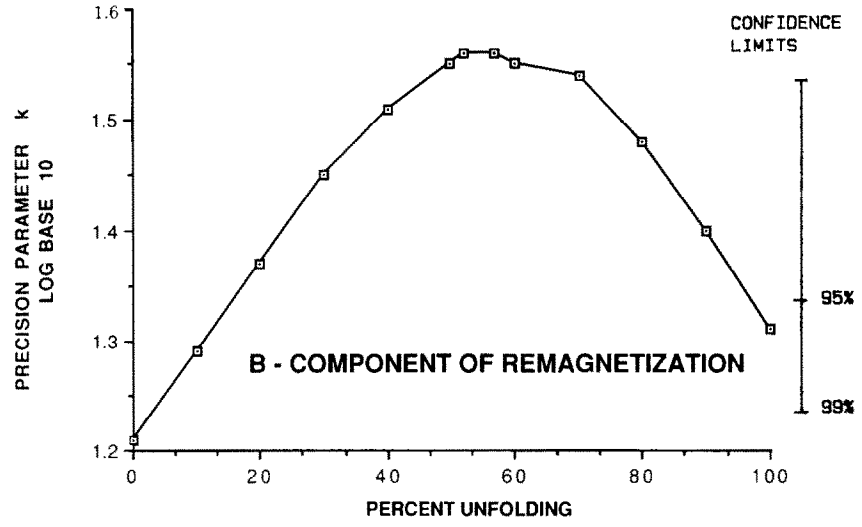


Fig. 10. Logarithm of Fisher's (1953) precision parameter (k) values versus percent unfolding for the mean directions derived from sites 106 and 107a from Junggar. The 95% and 99% confidence limits (vertical line) are also plotted.

The incremental fold test on these two sites suggests that the B component was imposed at 50% unfolding, and the precision parameter k at this unfolding step is greater than both 0% unfolding (geographic coordinates) and 100% unfolding (stratigraphic coordinates) at more than 95% confidence (Fig. 10).

The discovery of the same B component in both the Shanzhuquan and Kelamayi areas not only indicates that the remagnetization is of regional extent, but also puts critical constraints on the age and mechanism of the overprint because the granite in the Kelamayi area yielded two sets of radiometric ages. The Permian K–Ar age is critical in constraining the age of the overprint, which is discussed below.

Discussion of paleomagnetic results from Junggar Basin

As shown above, two well defined components (A and B) and one less well defined component (the higher unblocking temperature component in Late Permian sandstones) of remanent magnetization, neglecting the Quaternary weathering overprint carried by goethite in Jurassic mudstones, are present in rocks from western Junggar, and must be considered in order to draw tectonic conclusions from them.

The A component

In all sampling localities from Junggar, the A component is pervasively represented and clearly defined, regardless of rock age and lithology. This component fails both regional and local fold tests. It is, therefore, clearly a post-Jurassic overprint. The facts that this component is of normal polarity in 95% of the samples and that its mean direction lies within 3° of today's geocentric axial dipole direction suggest that it was mainly acquired in the last 0.7 Ma.

The B component

This component was found exclusively in Carboniferous rocks and was usually isolated only after removal of the A component. It is characterized in all cases by linear demagnetization vector segments and upward inclinations. However, the secondary nature of the B component is demonstrated by the failure of standard fold tests. Accepting the B component as secondary, a critical question then would be: when did the overprint occur and by what mechanism? Because the mean direction is southwest and up, it clearly does not represent the recent geomagnetic field direction for the Junggar region. The B component is apparently older than the A component based on

two facts: (1) the B component can be seen only in strata of Carboniferous age, while the A component is widespread in all sampled formations, and (2) the A component yields entirely post-folding results, while the B component appears to show synfolding results. The oldest folding that affected all sampled strata is of Permian age. Hence, the abundance of synfolding signal would suggest that the B component, at least in some strata, was imposed during deformation in the Permian.

The direction of the B component is always reversed, supporting the possibility of overprinting during the Late Paleozoic Kiaman Reversed Interval (Irving and Pullaiah, 1963; Khramov and Rodionov, 1981; Harland et al., 1982; Palmer, 1983; Opdyke, 1986). This would place boundaries of about 320–280 Ma for the remagnetization.

Another way of estimating the age of the B component is to consider the *reductio ad absurdum* argument for the age of Late Hercynian orogeny in the Junggar Basin. As mentioned previously, thermal activities peak during the Late Hercynian orogeny in western Junggar, and most published radiometric dates for igneous intrusions of this orogeny correspond to the Permian. The age of this orogenic even is also supported by the results of analysis of stratigraphy and plate motion (Yang and Yang, 1985; Wang and Liu, 1986). On the other hand, the age of remagnetization determined by the synfolding magnetization is also within this time period. The coincidence of a regional remagnetization with the activities of the Late Hercynian orogeny not only supports a Permian age for the B component, but also leads us to suspect a thermal origin for the overprinting.

The reported K–Ar age from the Middle Carboniferous granitic bodies further constrains the age of overprinting. Because the K–Ar age is the time of cooling to the argon closure temperature, the yielded Permian age would suggest a tectono-thermal event close to Permian time, perhaps associated with the Late Hercynian orogenic phases that not only wiped out the original remanent magnetization of the granite, but reset their K–Ar mineral ages as well.

Because the facts that the B component has a very narrow and uniform unblocking temperature (550–600 °C) range and a single reversed polarity, we believe that a thermal process is the most likely cause for the overprint. It is interesting to note that widespread overprinting of Paleozoic rocks in Tarim during the Permian has recently been reported (Zhai et al., 1987, 1988; Sharps et al., 1988). Because all geological estimates on the convergence of the Tarim and Junggar blocks point to a period of Late Carboniferous to Permian time (Zhang et al., 1984; Zhu, 1984; Coleman, 1989; Feng et al., 1989), it appears that regional thermal remagnetization of Junggar (the B component) and Tarim was caused by the phase of the Late Hercynian orogeny associated with the accretion of Tarim to Junggar. The widespread occurrence of A-type granitic intrusions in this region has also been interpreted as evidence that a huge Pan-Asian thermal event took place during and after the consolidation of central Asia (Coleman, 1989).

The close spatial and temporal connections between the accretionary history and regional remagnetization of these two blocks provide another example of a tectono-thermal overprinting even in a continental collision zone, and re-emphasize the problem of remagnetization associated with orogeny. Remagnetization events associated with Late Paleozoic orogeny and regional uplift in various parts of the world are now recognized as a rather frequent occurrence. Numerous examples have been well documented in Europe (Creer, 1968; Courtillot et al., 1986) and in North America (Gillett and Van Alstine, 1982; Irving and Strong, 1985; Van der Voo, 1989), as well as in Australia (Schmidt et al., 1987). Many Early and Middle Paleozoic paleomagnetic poles from the Russian platform may also have been remagnetized during the Late Paleozoic (Khramov and Rodionov, 1981).

The C component

This component was only found in Late Permian sandstone of the Liusugou area, and is less well defined than both A and B components. The mean direction of the C component is interesting for two reasons. First, this component is isolated after removal of the A component. As discussed

previously, because the C component is absent in interbedded conglomerates that do carry the A component, the C component might be primary. Secondly, this component has a reversed polarity in every case, which is consistent with the Kiaman Long Reversal Superchron. However, owing to several uncertainties (the effects of strain in redbeds, unrecognized plunging fold axes, or possible rotations due to the nearby Darbute fault, see Fig. 1c), no data at present can confidently be regarded as indicative of a primary Late Permian component direction. Our best guess at this stage is that the C component may also be overprinted in the Permian. Because the data are too scattered, we perfer not to include them in the final average of the regional remagnetization of Junggar.

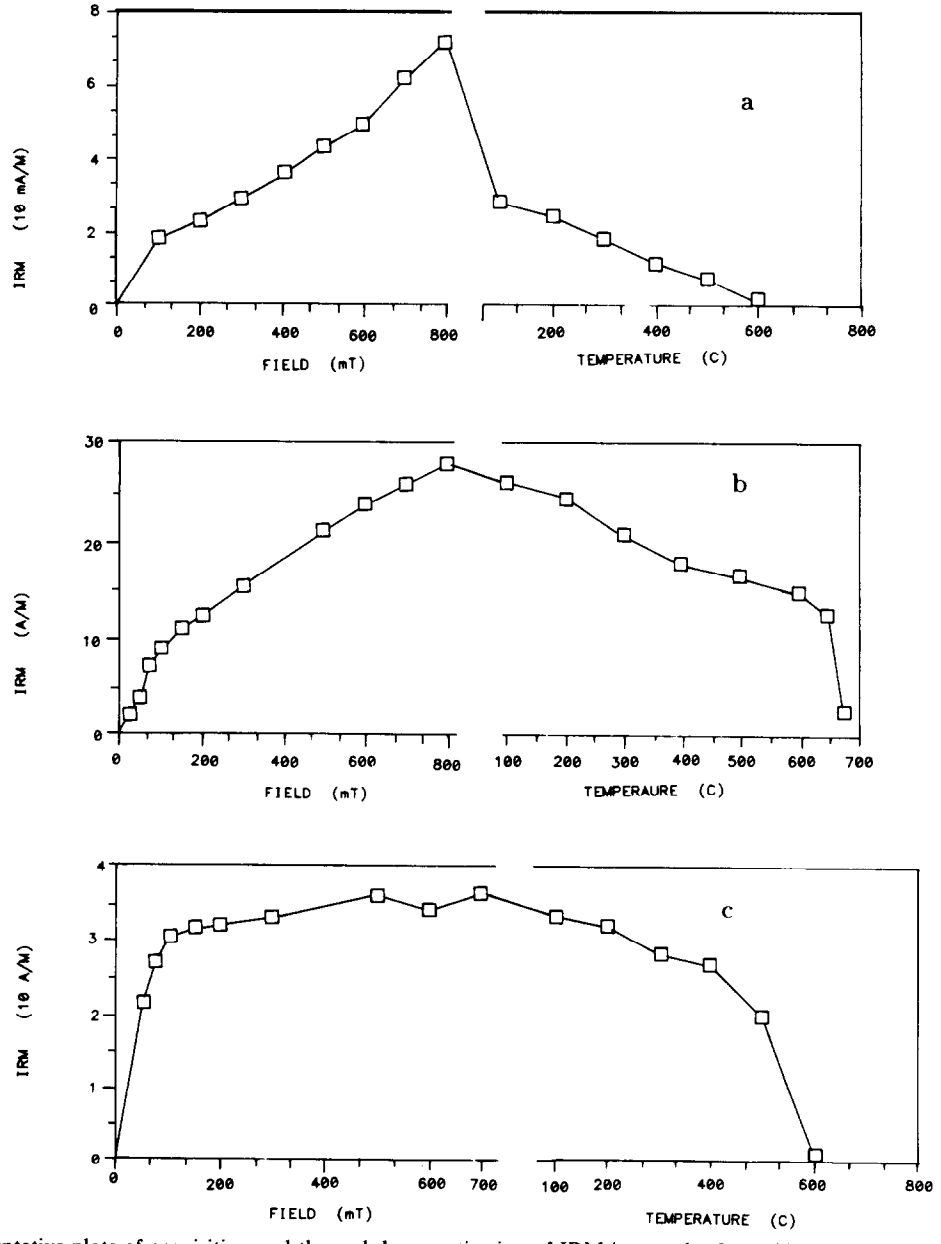


Fig. 11. Representative plots of acquisition and thermal demagnetization of IRM in samples from this study. (a) Type 1—magnetic properties characterized by goethite and magnetite. (b) Type 2—magnetic properties characterized by hematite and probably some magnetite. (c) Type 3—magnetic properties characterized by magnetite.

IRM experiments on samples from Inner Mongolia and Junggar

In order to complement AF and thermal treatments, we have chosen some representative samples from Inner Mongolia and Junggar for acquisition of isothermal remanent magnetization studies. More information about magnetic minerals can be obtained from observations of the decay of saturation isothermal remanence in rock samples heated in a field-free space than of thermal demagnetization alone. The identification of magnetic mineralogy is based on the fact that the decay curves of IRM during heating reflect the spectrum of unblocking temperatures in a sample and are characterized by distinct changes in slope.

As shown in Fig. 11, the spectrum of the IRM properties of rocks from areas sampled in this study can be grouped into three types (1–3). In the type 1 curve, the break in slope in the IRM acquisition curve around 100 mT, and the absence of blocking temperatures above 600 °C, are consistent with the presence of magnetite, whereas the failure to saturate IRM even by 800 mT and the sharp decrease in IRM after demagnetization to only 100 °C are characteristic of goethite. This behavior was shown mainly by Middle Jurassic sandstones from Junggar. The type 2 curve (predominantly in redbeds and reddish-colored andesites from Inner Mongolia, but also in redbeds from Junggar), domonstrates the presence of hematite by its high coercivity (> 800 mT) and high blocking temperatures (up to 680 °C), and also suggests the presence of a phase such as magnetite by the break in slope in IRM acquistion around 100 mT. Type 3 behavior, which is more commonly observed in rocks from both regions, is evidence for the presence of magnetite, with saturation IRM achieved by 200 mT and maximum unblocking temperatures ranging from 550 to 600 °C.

Tectonic implications

Table 5 summarizes the available paleomagnetic poles that have been obtained from North China, Inner Mongolia, Outer Mongolia, Siberia, Tarim, Junggar, Kazakhstan, Russia, and Europe. A detailed discussion of these paleomagnetic poles (Late Carboniferous onward) is presented elsewhere (Zhao and Coe, manuscript in preparation, 1989). The Carboniferous poles for Tarim, Kazakhstan, Russia, Europe, and Siberia are in rough agreement, suggesting that these regions may have been situated near each other and could have started the process of accretion in Late Carboniferous time. The presence of shallow marine and littoral sediments in both the Junggar and the Irtysh–Zaysan basins further suggests that the ocean had become restricted or nearly closed in these areas by the beginning of the Late Carboniferous (Zonenshain, 1973; Mossakovskiy and Dergunov, 1983; Feng et al., 1989). On the other hand, the preliminary Carboniferous paleomagnetic poles for Inner Mongolia, Outer Mongolia, and North China are distinct at much greater than 95% confidence (Table 5 and Fig. 12), indicating that these continental blocks could not have been connected in their present configuration during Carboniferous time.

As shown in Fig. 13a, the Late Permian paleomagnetic poles for the various crustal blocks of Asia in Table 5 fall into the same two groups, which again are completely different from each other. The pole for Outer Mongolia is a little

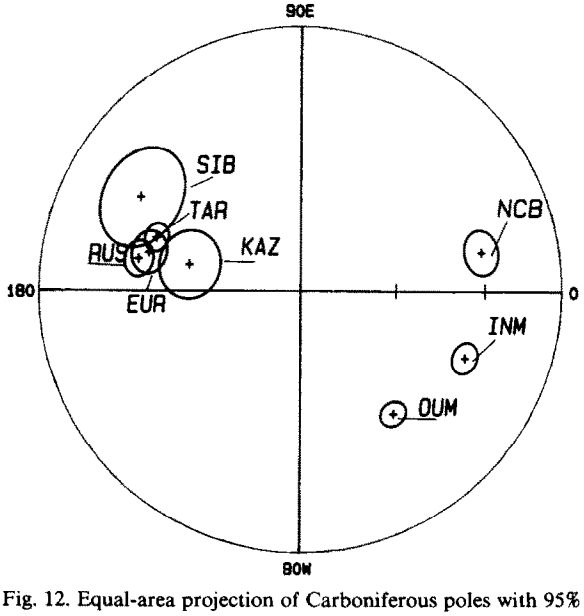


Fig. 12. Equal-area projection of Carboniferous poles with 95% confidence circles for various blocks in eastern Asia. Abbreviations same as Fig. 1.

TABLE 5

Paleopoles for Eurasian blocks

Block	Age	N	Long.	Lat.	A_{95} α_{95}	Reference
Mongolia–North China (MON–NC)						
INM	T	13	29.2	88.6	8.2	this study
INM	T	5	324.6	85.0	6.1	Zheng et al., 1989
Mean INM	T	18	351.2	87.9	6.0	average of the above two studies
INM	K	6	249.5	82.9	5.7	this study
OUM	K	3	241.7	82.9	(4.2)	Pruner, 1987
NCB	K	?	279.4	80.3	(7.2)	Xiong, 1986
	K	?	358.6	87.9	(3.5)	Tan, 1988
	K	?	310.2	84.6	(4.2)	Cheng et al., 1988
Mean NCB	k	3	297.6	84.8	7.4	average of the above 3 studies
Mean MON–NC	K	3	258.9	84.0	5.0	average poles for INM, OUM, and NCB
INM	J3	4	231.6	68.5	9.5	this study
NCB	J2	12	214.0	76.1	(12.8)	Fang et al., 1988
NCB	J2	4	205.5	77.0	6.1	Zheng et al., 1989
NCB	T3	4	20.5	62.4	(14.6)	Wu, 1988
NCB	T2	?	12.5	49.1	(3.9)	Cheng et al., 1988
	T2	?	18.3	47.6	(3.1)	Fang et al., 1988
	T2	?	19.9	51.9	7.7	Wu, 1988
Mean NCB	T2	3	16.9	49.6	5.1	average of the above 3 studies
NCB	T1	5	352.7	56.8	6	Tan, 1988
	T1	33	3.5	60.3	2	Tan, 1988
	T1	5	348.9	67.7	(3.6)	Wu, 1988
	T1	7	6.8	45.4	7	Lin, 1984
	T1	56	354.1	55.5	2	Zhang, 1984
Mean NCB	T1	5	358.1	57.3	8.6	average of the above 5 studies
OUM	P2	3	11.8	33.8	(8.5)	Pruner, 1987
INM	P2	11	339.5	52.5	9	this study
	P2	50	355.6	56.7	2.3	Liu, 1987
Mean INM	P2	12	342.4	52.5	6.4	average of the above 2 studies
NCB	P2	3	13.7	42.3	9	Cheng et al., 1988
	P2	12	1.4	55.2	6.3	Zhao and Coe, 1989
	P2	33	2.9	37.9	6.9	Zhao and Coe, 1989
	P2	19	335.7	55.2	4.7	Zhao and Coe, 1989
*	P2	12	178.5	−44.0	3.9	McElhinny et al., 1981
*	P2	82	178.5	−49.2	2	Zhang, 1984
*	P2	10	186.3	−38.1	?	Lin, 1984
*	P2	8	6.2	56.7	11.5	W. Lin et al., 1984 (unpublished data)
Mean *	P2	4	2.3	47.1	9.6	average 4 studies at Taiyuan, marked by asterisk (*)
	P2	30	171.3	−43.9	4.1	Coe et al., 1987
Mean NCB	P2	6	358.7	47.5	9.2	averaging mean * and other 5 studies
Mean MON-NC	P2	3	359.2	45.2	21.9	averaged by blocks
		9	358.3	47.8	7.7	averaged by studies (NCB, 6; INM, 2; OUM, 1)
NCB	C3	3	11.9	30.0	6.1	Wu, 1988
INM	C3	29	338.1	32.6	4.4	this study
OUM	C3	3	307.4	40.9	(3.9)	Pruner, 1987
Europe–Siberia–Kazakhstan (EUR–SIB–KAZ)						
SIB	K	7	180 [a]	74	10	Khramov, 1982
SIB	K	?	168 [a]	68	9	Khramov et al., 1985
EUR	K1	9	166 [a]	70	4	Westphal et al., 1986
EUR	K2	3	152 [a]	74	13	Westphal et al., 1986

TABLE 5 (continued)

Block	Age	N	Long.	Lat.	A_{95} α_{95}	Reference
Europe–Siberia–Kazakhstan (EUR–SIB–KAZ)						
EUR	K1	14	164 [a]	72	5	Lee et al., 1987
EUR	K1	15	153 [a]	70	6	Irving, 1977
TAR	K2	4	224	66	9	Li, 1990
S. France	K1	?	220	67	9	J. Besse and Boisseau; see Lee et al., 1987
W. Germany	K2	?	181	76	4	Heller and Channell, 1979
EUR	K2	(T-AFR)	184	81	7	Westphal et al., 1986
EUR	K1	(T-NA [b])	211	69	7	Westphal et al., 1986
EUR	K1	(T-AFR)	215	72	4	Besse and Courtillot, 1988
EUR	K1	(T-SA)	206	71	9	Westphal et al., 1986
EUR	K1	(T-NA)	221	73	6	Van der Voo and French, 1974
Mean	K	8	207.2	72.2	4.5	average of the above 8 studies
SIB	J3	?	125 [d]	51	?	Khramov et al., 1985
EUR	J3	(T-NA [b])	194	66	6	Westphal et al., 1986
EUR	J3	(T – AFR)	212	76	–	Westphal et al., 1986
EUR	J3 (140 Ma)	(T-AFR)	197	71	12	Besse and Courtillot, 1988
TAR	J3-K1	6	209	65	9	Li, 1990
Mean	J3	4	206.5	69.9	7.0	average of the above 4 studies
SIB	T2-3	?	137 [a]	59	8	Khramov et al., 1985
SIB	T1	20	147	52	4	Khramov and Rodinov, 1981
SIB	T1	?	149 [a]	46	4	Westphal et al., 1986 (not independent data)
TAR	T1	6	183	69	11	McFadden et al., 1988
EUR	T1	?	153	48	3	Irving and Irving, 1982
EUR	T1	?	152	49	3	Westphal et al., 1986
RUS	T1	11	156	53	5	Khramov and Rodinov, 1981
EUR	T1	(T-NA [b])	147	57	7	Westphal et al., 1986
KAZ	T1	18	142	44	13 [c]	Piper, 1988
Mean	T1	6	149.5	48.8	4	average of the above 6 studies and omitted TAR
SIB	P2	7	145	52	8	Khramov and Rodinov, 1981
KAZ	P2	55	169	54	9 [c]	Khramov and Rodinov, 1981
TAR	P2	4	183	65	7.7	averaging McFadden et al., 1988, and Li et al., 1988
WJB	P	11	178	51	6	this study
EUR	P2	3	152	48	14	Westphal et al., 1986
EUR(T-NA + SA)	P2	4	155	51	10	Westphal et al., 1986
Mean 1	P2	4	155.0	51.6	7.6	average of the above 4 studies and omitted TAR and WJB
SIB	C	4	150	30	14	Khramov and Rodinov, 1981
KAZ	C3	4	167	54	10	Khramov et al., 1981
EUR	C	14	169	37	5	Irving and Irving, 1982
RUS	C	18	166	40	6	Khramov and Rodinov, 1981
TAR	C3	3	160	41	4	Zhai et al., 1988

Abbreviations: SIB = Siberia, KAZ = Kazakhstan, TAR = Tarim, RUS = Russia, WJB = West Junggar Basin, EUR = Europe, NCB = North China Block, OUM = Outer Mongolia, INM = Inner Mongolia. (T-NA) = transferred data from North America, (T-SA) = transferred data from South America. (T-AFR) = transferred data from Africa.

[a] Data are not used in the reconstrction.

[b] Transferred from North America poles of Harrison and Lindh (1982).

[c] * Estimated from the reported *dp* value.

discrepant, but it is based only on the results of a preliminary study (Pruner, 1987). We will term the amalgam of the NCB and Inner and Outer Mongolia the Mongolia–North China plate (MON–NC). It is clear now from the paleomagnetic data that the Late Permian suturing referred to previously on geological evidence as being between the NCB and SIB along the Central Asian Fold Belt (Fig. 1) merely reflects the earlier process of accretion of the various terranes and blocks that make up the MON–NC plate. The final suture to close between the MON–NC plate and SIB lies further north in the Southern Siberian Fold Belt, which separates Mongolia and Siberia (Fig. 1a).

The Late Permian poles of Siberia, Europe, Kazakhstan, Tarim, and the Permian pole of west Junggar are similar to each other to a first approximation (Fig. 13a); however, there is a small but statistically significant difference between those for Tarim, Junggar, and Siberia, suggesting a relative motion between these blocks. For this reason, Tarim and Junggar were not included in the mean pole calculations (Table 5) and their tectonic relationship with respect to Siberia is discussed separately. The amalgamation of Siberia, Europe, and Kazakhstan is hereafter referred to collectively as the Europe–Siberia–Kazakhstan (EUR–SIB–KAZ) plate or, more simply, as Eurasia.

The mean Late Permian paleomagnetic pole positions for the MON–NC and EUR–SIB–KAZ plates are plotted in Fig. 13b. The great circle that is equidistant between the paleomagnetic poles for the two plates almost intersects the Southern Siberian Fold Belt near its western end (the Irtysh–Zaysan Basin). The two paleomagnetic poles can be brought into coincidence by an Euler rotation about a pole lying anywhere on this great circle. Because a suture may be either a compression line (as in a subduction zone) separating two plates, or an ancient transform fault, we will consider two simple cases in this paper for the two alternative boundaries. The first case is a transform boundary with an Euler pole near the north pole, involving a gigantic east–west fault system lying within the approximately east–west trending Southern Siberian Fold Belt. Displacements that

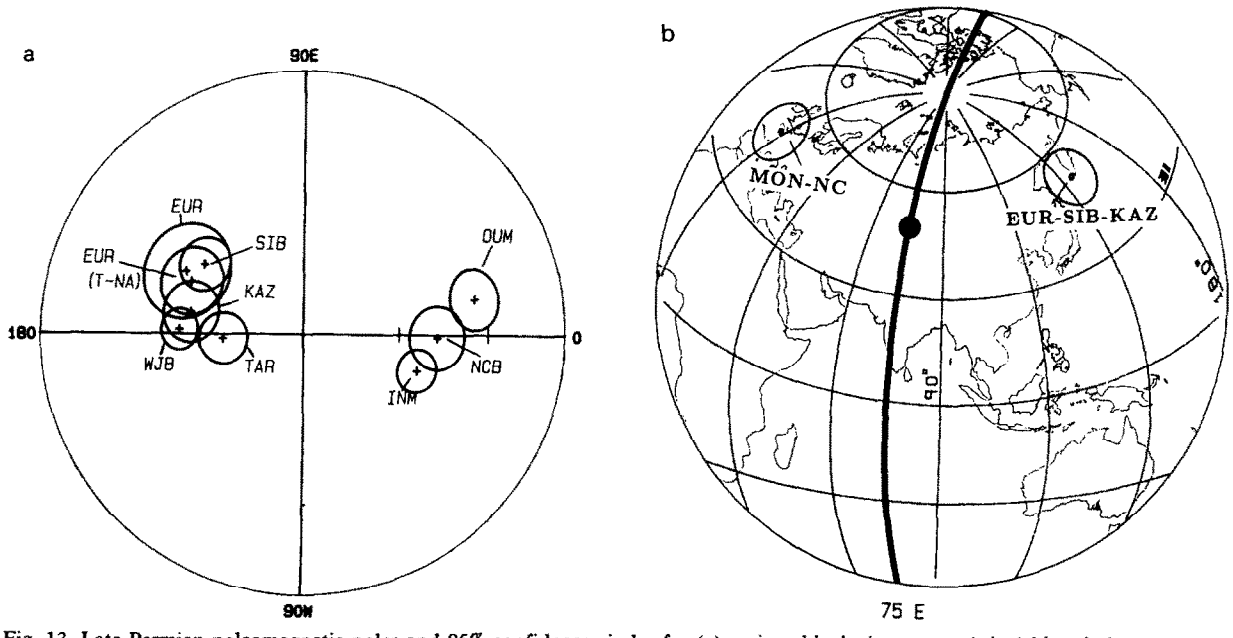


Fig. 13. Late Permian paleomagnetic poles and 95% confidence circles for (a) various blocks in eastern Asia (abbreviations same as Fig. 1) and (b) the MON–NC and EUR–SIB–KAZ plates. Heavy line in (b) near longitude 75° E is the bisecting great circle along which lie the Euler poles that could rotate the two poles into coincidence. The large black dot represents the Euler pole at 50° N, 75° E favored in this study.

are needed to produce the present separation of the Late Permian poles for the MON–NC and EUR–SIB–KAZ plates, are prohibitively large: 11,000 km of left-lateral displacement, or 14,500 km of right-lateral displacement.

The second case entails subduction between the MON–NC and EUR–SIB–KAZ plates along the Southern Siberian Fold Belt. A rotation of 117° about an Euler pole at the intersection (50° N, 75° E) between the great circle and the fold belt

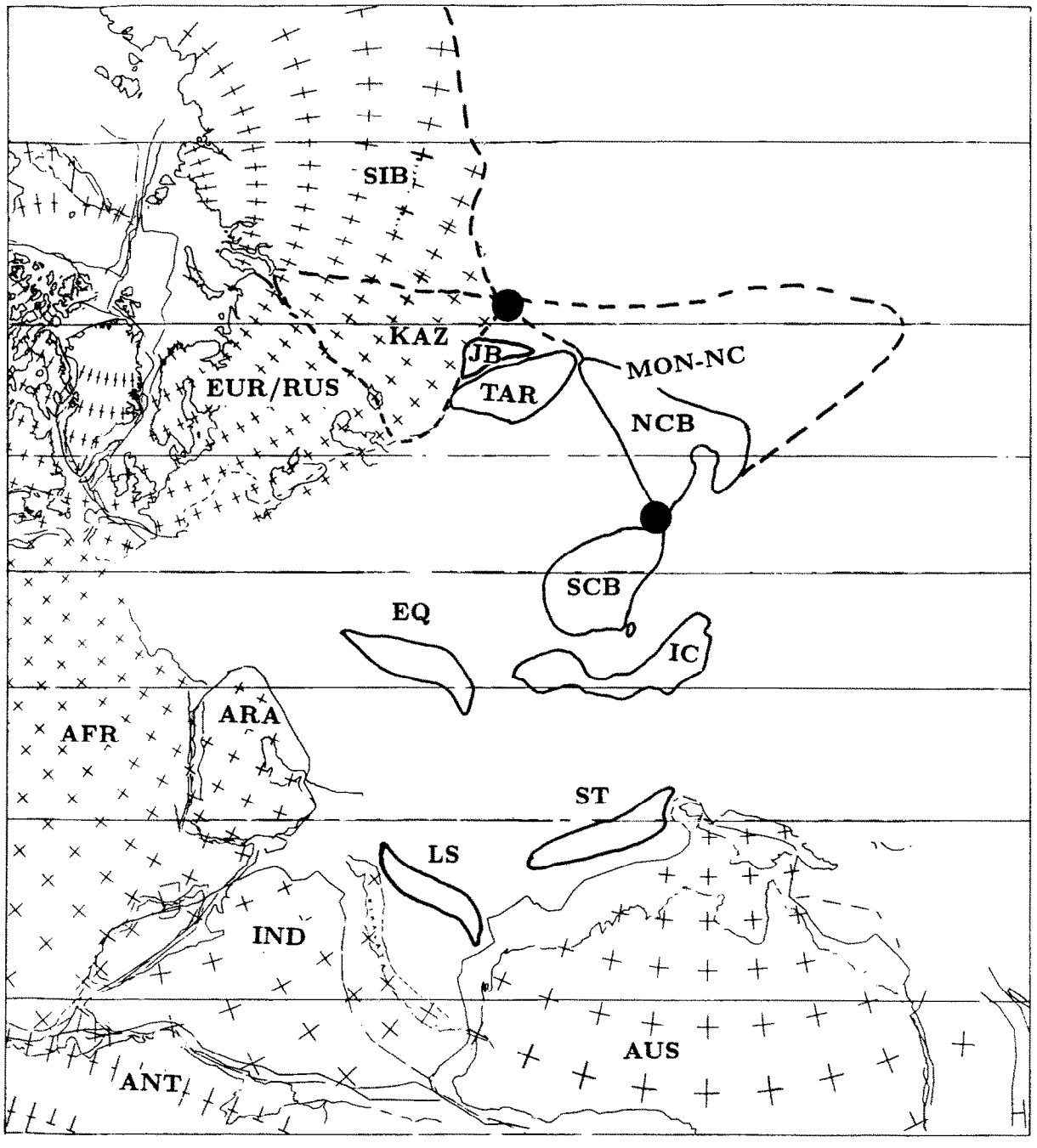


Fig. 14. Schematic reconstruction of Mongolia–North China (MON–NC), Europe–Siberia–Kazakhstan (EUR–SIB–KAZ), Tarim (TAR), Junggar Basin (JB), and the South China Block (SCB) in Late Permian time. The positions of Africa (AFR), Arabian Shield (ARA), India (IND), Australia (AUS), Antarctica (ANT), the East Qiangtang (EQ), Lhasa (LS), Indo-China (IC), and Shan-Thai (ST) blocks modified after Nie et al. (1989); Mercator projection.

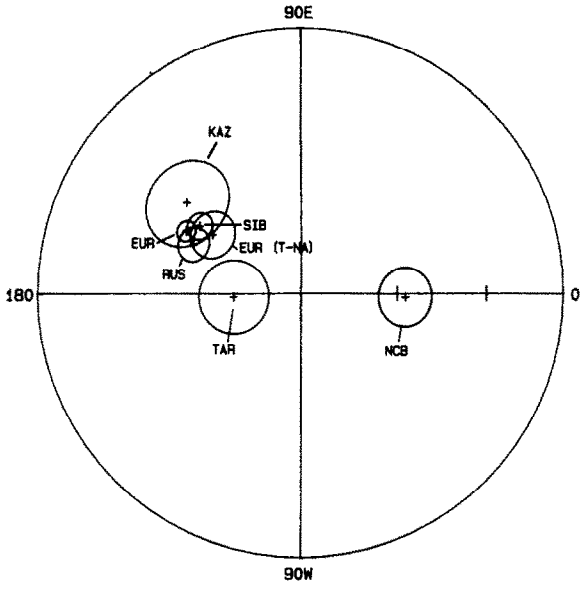


Fig. 15. Equal-area projection of Early Triassic mean paleomagnetic poles for blocks of the MON–NC and EUR–SIB–KAZ plates.

brings the poles into coincidence. As shown in Fig. 14, collision might have occurred initially at the western and in the Irtysh–Zaysan region near the east edge of the Kazakhstan block (in today's geographic frame) and progressed eastward as the MON–NC plate rotated counterclockwise relative to Eurasia.

Early Triassic poles (Table 5 and Fig. 15) for all continental blocks (with the exception of Tarim) are very similar to those for the Late Permian, consistent with the same model just discussed in relation to the Late Permian data. In this case, a total of 111 degrees rotation about an Euler pole at lat. = 50° N, long. = 78° E is required to reconcile the Early Triassic poles. The bisecting great circle nearly cuts through the Euler pole for the Late Permian case, and the amount of relative rotation is only 5.5° less.

Middle to Late Triassic and Middle Jurassic paleomagnetic data are sparse and are only available for the North China Block (Table 5). Although these data are still of a preliminary nature, the pattern of counterclockwise rotation of the NCB through these time periods seems to emerge. The next coeval paleomagnetic results available from both plates are Late Jurassic. As shown in Table 5 and Fig. 16, the Late Jurassic paleomagnetic poles for the MON–NC and EUR–SIB–KAZ plates statistically indistinguishable. Therefore, large relative motion between the two plates was mainly accomplished sometime between Early Triassic and Late Jurassic time, quite possibly during the Early Jurassic. Cretaceous paleomagnetic poles for the two plates do no entirely coincide with each other, indicating that a small amount of relative motion between the two plates. This post-Cretaceous movement was likely caused by the collision of Indian with Tibet (Li, 1990).

Several geological and geophysical observations favor the second (subduction) model. The first and most compelling observation is the distribution of marine sediments and alkalic granites within the collision zone. The abrupt regional and local changes from marine to continental basin deposition and the occurrence of syncollisional and postcollisional alkalic granites during the Carboniferous and Permian in the West Junggar Basin of northwest China, the Irtysh–Zaysan basin of the U.S.S.R., and the Southern Mongolian Basin of the Mongolian People's Republic (Mossakovskiy and Dergunov, 1983; Karyayev, 1984; Kurchavov and Yarmolyuk, 1984; Coleman, 1989; Feng et al., 1989) indicate the withdrawal of the Paleozoic ocean and signal a subduction rather

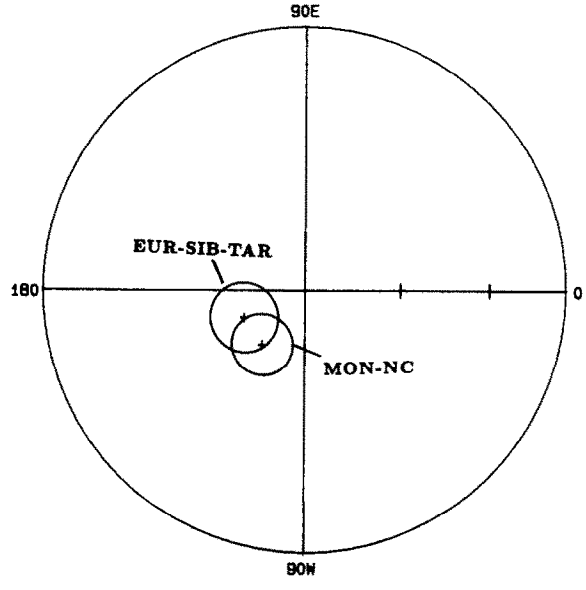


Fig. 16. Equal-area projection of Late Jurassic mean paleomagnetic poles for the MON–NC and Eurasia (EUR, SIB, and TAR data) plates.

than a transform boundary between the MON–NC and the EUR–SIB–KAZ plates. Both Nalivkin (1973) and Kosygin and Parfenov (1975, 1981) reported that Triassic marine deposits are widespread in basins of the Southern Siberian Fold Belt to the east, but that only continental deposits are developed in the western part of this fold belt (Altay–Sayan region) during this time. Coal deposits ceased completely in the west by Permian time, but continued to exist until Late Jurassic time in the east (Tamrazyan, 1971). Moreover, Kosygin and Parfenov (1981) reported that continuous marine fossils exist from Cambrian to Mid-Jurassic time in the Mongolo–Okhotsk eugeosynclinal system (i.e., developed through oceanic, transitional, and continental stages) located near the east end of the South Siberian Fold Belt. Finally, the known Hercynian orogenic phases are thought to record convergence and collision in Late Carboniferous to Permian time, and the paired high- and moderate-pressure metamorphic belts in the Zaysan region appear to mark the site of a subduction suture (Yermolov et al., 1984). All these geological observations are readily explained if an ocean between the two plates had closed in the west due to collision in the Late Permian, but remained open to the east until Late Jurassic time, as shown in Fig. 14.

The paleontologic data are also in good agreement with the paleomagnetic results obtained from this study and are counter to the transform boundary model. Permian fossil plants found in Tarim show a mixture of Angaran (north) and Cathaysian (south) floras (Wu, 1983), while in Permian strata of Inner and Outer Mongolia to the east, only Cathaysian-type flora has been discovered (Huang, 1986). Carboniferous flora from the Primorye block of the southeastern portion of the Soviet Union are different from those of Mongolia (Nalivkin, 1973; Khain, 1985). This is consistent with a large ocean barrier existing in the east between the MON–NC and Siberia, but not to the west in Tarim and Junggar.

The distribution of igneous activity fits the sense of rotation in our model very well. Syncollisional and postcollisional A-type granites and calc-alkalic volcanic rocks in the collision zone are widespread. They are of Late Paleozoic age in the west (Karyayev, 1984; Zonenshain et al., 1985; Feng et al., 1989; Kwon et al., 1989), but mainly of Mesozoic age in the east (Zonenshain et al., 1985, p. 105, fig. 2; Kosygin and Parfenov, 1981, p. 397). This pattern would coincide with a late Paleozoic convergence in the west and Mesozoic consolidation in the east by relative rotation of the two plates. No evidence of igneous activity has been identified in the west Junggar basin and Tarim since the Late Paleozoic thermal event, but the widespread invasion of northeastern Inner Mongolia by Late Jurassic andesitic and tuff flows is striking. These thermal events are especially important because they not only have remagnetized the older rocks in this region, but also probably mark the termination of convergence between the MON–NC and EUR–SIB–KAZ plates. Moreover, because of the existence of Permo-Triassic alkalic granites and salt-bearing lagoonal and coarse red molasse formations, which indicate an active continental margin, and high- (kyanite–sillimanite) and moderate- (granulite–amphibolite) pressure metamorphic belts in the Irtych–Zaysan region during the end of the Hercynian orogeny (i.e., Permo-Triassic), and the later, widespread eruption of andesitic rocks in Inner and Outer Mongolia during the Late Jurassic, it appears that a north-dipping subduction zone existed in the west in Permo-Triassic time, whereas in the east during Late Jurassic time subduction was south-dipping.

Last, but not least, some geophysical observations available from the studied areas are also relevant to our model. High positive magnetic anomalies associated with the Mongolo–Okhotsk basin at the east end of the boundary suggest that there is mafic material in the basement, probably accreted oceanic crust (Kosygin and Parfenov, 1981). Based on gravity measurements, the crust of the Junggar basin is rather thin, around 32–34 km, whereas the crust in the Irtysh–Zaysan is ~ 50 km thick (Coleman, 1989). This might indicate that collision indeed occurred in the Irtysh–Zaysan region.

As mentioned previously, there is a small but significant difference between the Late Permian poles for Siberia and Tarim, indicating a relative motion between the two blocks. The Late Permian

pole for Tarim is offset northeast relative to Siberia (Fig. 13a). The simplest explanation is that Tarim moved northeast since the Permian (Sharps et al., 1989; Li, 1990). We found that a 30° counterclockwise rotation of Tarim with respect to Siberia about an Euler pole located at 60°N latitute, 45°E longitude can account for the difference between their Late Permian poles. According to this reconstruction, Tarim moved northeast about 1400 km relative to Siberia along a transform fault that corresponds approximately to the boundary of the Kazakhstan block.

The Early Triassic poles for Siberia and Tarim (Table 5 and Fig. 15) are very similar to those for the Late Permian, indicating that the relative northeastward motion occurred after the Early Triassic. As shown in Fig. 16, the Late Jurassic poles for Siberia and Tarim overlap each other, suggesting the motion was complete by this time. Several left-lateral strike-slip faults have been documented along the boundary between the Kazakhstan and Tarim blocks (Ni, 1978; Molnar and Deng, 1984; Carroll et al., 1876; Feng et al., 1989). We speculate that Tarim may initially have collided with Kazakhstan in late Paleozoic time, then moved northeastward along strike-slip faults and docked with respect to Eurasia by Late Jurassic time.

It perhaps should be pointed out that this Late Jurassic docking time is different from the interpretation by Li (1990). Using the Cretaceous data from Siberia, Li concludes that the relative motion occurred after the Cretaceous. However, the terminal stage of the relative motion depends on whether we compare the Tarim data with the Siberia data or the data transferred from other continents. We believe the transferred data are better for Jura-Cretaceous time for the following reasons. Several other workers in addition to ourselves have suggested that the Mesozoic Siberia data may be in serious error (Frei and Cox, 1987; Lee et al., 1987, p. 3580; S.T. Kwon and R. Van der Voo, pers. commun., 1989). On the other hand, the transferred poles for Europe agree well with two new Cretaceous poles of J. Besse and Boisseau (unpublished data, 1986, see Lee et al., 1987, p. 3593) and Heller and Channell (1979), and together they give the best Cretaceous pole for Europe. Under the reasonable and widely accepted assumption that Europe and Siberia were a stable tectonic unit long before Cretaceous time, this pole should apply equally well for Siberia. In support of this, we note that the transferred Late Permian and Early Triassic poles agree well not with each other but also with the coeval poles from Europe and Siberia (Table 5). For all these reasons, and because the Cretaceous poles for Europe and Tarim are consistent (Table 5), we conclude that the northeastward motion of Tarim relative to both Siberia and Europe occurred mainly between Early Triassic and Late Jurassic time.

Conclusions

Paleomagnetic evidence supports the hypothesis that the Tarim, Kazakhstan, Russia, Europe, and Siberia blocks were situated near each other and could have started the process of accretion in Late Carboniferous time. The preliminary Carboniferous paleomagnetic poles for Inner and Outer Mongolia and North China, on the other hand, are significantly different from the abovementioned Eurasia blocks at more than 95% confidence, indicating that these regions could not have been connected to Eurasia in their present configuration during Carboniferous time.

Late Permian paleomagnetic results suggest that Kazakhstan, Russia, Europe, and Siberia were amalgamated by this time to form the EUR–SIB–KAZ (Eurasia) plate. Outer and Inner Mongolia and the North China Block were also amalgamated with each other by the Late Permian, forming the MON–NC plate. These two plates were not a stable unit during the Late Permian. The geometrical relationship of their paleomagnetic poles suggests that continental collision initially occurred around Permo-Triassic time near the western end (50°C, 75°E) of the South Siberian Fold Belt, and suturing continued eastward as the MON–NC plate rotated counterclockwise about 117° relative to the EUR–SIB–KAZ plate during Early Mesozoic time. Eastward younging in igneous activity and westward disappearance of Jurassic marine sediments on the northern margin of the MON–NC plate support this model. Sutur-

ing and rotation were essentially complete by the Late Jurassic, and perhaps as early as Early to Middle Jurassic.

As mentioned in the Introduction, accretion of the NCB and SCB also took place sometime between Late Permian and Cretaceous time (Zhao and Coe, 1989), possibly by a mechanism of collision and subsequent rotation similar to that suggested above for the MON–NC and EUR–SIB–KAZ plates, but antithetic in sense of rotation (Fig. 14). This kind of mechanism (i.e., initial collision in a narrow zone and subsequent suturing by rotation around a pole near the collision zone) may be a common tectonic style in the juxtaposition of large blocks of continental crust.

Late Permian and Early Triassic poles for Tarim are offset to the northeast relative to Siberia, suggesting that significant relative motion between the two blocks occurred after the Early Triassic. This motion is characterized by 1400 km of northeastward movement of Tarim relative to Siberia along a left-lateral strike-slip fault that corresponds approximately to the southeastern boundary of the Kazakhstan block, and ended essentially by the Late Jurassic.

Thus, our own and others' paleomagnetic studies indicate that the Late Triassic and Early Jurassic are the key periods for the consolidation of Chinese continents. Paleomagnetic data from the Chinese blocks during these key periods are needed to test and refine the models shown in Fig. 14 and to better constrain the tectonic history of China.

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References

Bai, Y., Chen, G., Sun, Q., Li, Y.A., Dong, Y. and Sun, D., 1987. Late Paleozoic polar wander path for the Tarim platform and its tectonic significance. Tectonophysics, 139: 145–153.

Besse, J. and Courtillot, V., 1988. Paleogeographic maps of the continents bordering the Indian Ocean since the early Jurassic. J. Geophys. Res., 93: 11791–11808.

Carroll, A.R., Graham, S.A., McKnight, C.M., Xiao, Z. and Liang, Y., 1987. Paleozoic tectonics and sedimentation, northern Xinjiang Autonomous Region, China. EOS, Trans. Am. Geophys. Union, 68: 1444 (abstract).

Chen, Z. and Liang, Y., 1985. Research on several problems of structural geology in Tianshan, Xinjiang. Xinjiang Geol., 3: 1–13 (in Chinese with English abstract).

Cheng, G.L., Bai, Y.H. and Sun, Y.H., 1988. Paleomagnetic study on the tectonic evolution of the Ordos Block, North China. Seismol. Geol., 10: 81–87 (in Chinese with English abstract).

Chinese Academy of Geological Sciences, 1973. The Geological Atlas of People's Republic of China Cartographic Publishing House, Beijing, 149 pp. (in Chinese).

Coe, R., Globerman, B., Plumley, P. and Thrupp, G., 1985. Paleomagnetic results from Alaska and their tectonic implications. In: D.G. Howell (Editor), Tectonostratigraphic Terranes of the Circum-Pacific Region. Am. Assoc. Pet. Geol., Circum-Pac. Counc. Energy Resour., Earth Sci. Ser.,: 85–108.

Coe, R., Zhao, X. and Meng, Z., 1987. Tectonic implications of paleomagnetic results from North and South China. Abstr., Int. Union Geod. Geophys., 1: 115.

Coe, R., Zhao, X., Zhou, Y., Li, S., Gao, Y., Ma, X. and Lin, Y., 1988. Paleomagnetism of Some Late Paleozoic and Mesozoic Rock Units From Inner Mongolia, China. EOS, Trans. Am. Geophys. Union, 69: 1171 (abstract).

Coleman, R.G., 1989. Continental growth of northwest China. Tectonics, 8: 621–635.

Courtillot, V., Chambon, P., Brun, J.P. and Rochette, P., 1986. A magnetotectonic study of the Hercynian Montagne Noire (France). Tectonics, 5: 733–751.

Cox, A., 1970. Latitude dependence of the angu'ar dispersion

of the geomagnetic field, Geophys. J. R. Astron. Soc., 20: 253–269.

Creer, K.M., 1968. Paleozoic paleomagnetism. Nature, 219: 246–250.

Deng, J.F., E, M.L. and Lu, F.X., 1988. The chemistry of Hanouba basalts and their trends of magmatic evolution. Acta Petrol. Sinica, 1: 22–33 (in Chinese with English abstract).

Fang, D.J., Guo, Y.B., Wang, Z.L., Tan, X.D., Fan, S.Q., Yuan, Y.R., Tang, X.Z. Wang, B.G., 1988. Tectonic Implication of Triassic and Jurassic Paleomagnetic Results from Ningwu Basin, Shanxi. Kexue Tongbo, 2: 133–135 (in Chinese).

Feng, Y., Coleman, R., Tilton, G. and Xiao, X., 1989. Tectonic evolution of the West Junggar Mountains, Xinjiang, China. Tectonics, 8: 729–752.

Fisher, R.A., 1953. Disperson on a sphere. Proc. R. Soc. London, Ser. A, 217: 295–305.

Frei, L.S. and Cox, A., 1987. Relative displacement between Eurasia and North America prior to the formation of oceanic crust in the North Atlantic. Tectonophysics, 142: 111–136.

Gillett, S.L. and Van Alstine, D.R., 1982. Remagnetization and tectonic rotation of Upper Precambrian and Lower Paleozoic strata from the desert range, southern Nevada. J. Geophys. Res., 87: 10929–10953.

Graham, J.W., 1949. The stability and significance of magnetism in sedimentary rocks. J. Geophys. Res., 54: 131–167.

Harland, W.B., Cox, A.V., Llewellyn, P.G., Pickton, C.A.G., Smith, A.G. and Walters, R., 1982. A Geologic Time Scale. Cambridge University Press, Cambridge, 131 pp.

Harrison, C.G.A. and Lindh, T., 1982. A polar wander curve for North America during the Mesozoic and Cenozoic. J. Geophys. Res., 87: 1903–1920.

Heller, F. and Channell, J.E.T., 1979. Paleomagnetism of Upper Cretaceous limestones from the Münster basin. J. Geophys., 46: 413–427.

Hsü, K.J., Wang, Q.C., Li, J.L., Zhou, D. and Shu, S., 1987. Tectonic evolution of Qinling Mountains, China. Eclogae Geol. Helv., 80: 735–752.

Huang, B.H., 1983. On late Paleozoic paleophytogeographic regions of eastern Tianshan–Hingan foldbelt and its geological significance. In: Contributions to the Project on Plate Tectonics of North China, 1: 138–155 (in Chinese with English abstract).

Huang, T.K., 1978. An outline of the tectonic characteristics of China. Eclogae Geol. Helv., 71: 611–635.

Il'in, A.V., 1986. Tectonics of south China. Geotectonics, 20: 21–30.

Irving, E. and Irving, G.A., 1982. Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana. Geophys. Surv., 5: 141–188.

Irving, E. and Parry, L.G., 1963. The magnetism of some Permian rocks from New South Wales. Geophys. J. R. Astron. Soc., 7: 395–411.

Irving, E. and Strong, D.F., 1985. Paleomagnetism of rocks from Burin peninsula, Newfoundland: hypothesis of Late Paleozoic displacement of Acadia criticized. J. Geophys. Res., 90: 1949–1962.

Ji, X. and Coney, P., 1985. Accreted terranes of China. In: D.G. Howell (Editor), Tectonostratigraphic Terranes of the Circum-Pacific Region. Am. Assoc. Pet. Geol., Circum-Pac. Counc. Energy Resourc., Earth Sci. Ser., 1: 349–361.

Karyayev, V.A., 1984. Evolution of the Paleozoides in Central Kazakhstan. Geotectonics, 18: 322–333.

Khramov, A.N., 1982. Paleomagnetism Leningrad, 303 pp. (in Russian).

Khramov, A.N. and Rodionov, V.P., 1981. The geomagnetic field during Palaeozoic. time. In: M.W. McElhinny, A.N. Khramov, M. Ozima and D.A. Valencio (Editors), Global Reconstruction and the Geomagnetic Field during the Palaeozoic. Adv. Earth Planet. Sci., 10: 99–115.

Khramov, A.N., Gurevich, E.L., Komissarova, R.A., Osipova, E.P., Pisarevsky, S.A., Rodionov, V.P. and Slautsitais, I.P., 1985. Paleomagnetism, microplates and Siberian plate consolidation. J. Geodyn., 2: 127–139.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc., 62: 699–718.

Klimetz, M.P.., 1983. Speculations on the Mesozoic plate tectonic evolution of eastern China. Tectonics, 2: 139–166.

Kosygin, Y.A. and Parfenov, L.M., 1975. Structural evolution of eastern Siberia and adjacent areas. Am. J. Sci., 275-A: 187–208.

Kosygin, Y.A. and Parfenov, L.M., 1981. Tectonics of the Soviet Far east. In: A.E.M. Nairn, M. Churkin and F.G. Stehli (Editors), The Ocean Basins and Margins, 5: 377–412.

Kurchavov, A.M. and Yarmolyuk, V.V., 1984. Distribution of Permian continental volcanics in Central Asia and its tectonic interpretation. Geotectonics, 18: 344–354.

Kwon, S.T., Tilton, G.R., Coleman, R.G. and Feng, Y., 1989. Isotopic investigations of the west Junggar region, Xinjiang, China. Tectonics, 8: 719–727.

Lee, G., Besse, J., Courtillot, V. and Montigny, R., 1987. Eastern Asia in the Cretaceous: New Paleomagnetic Data From South Korea and a New Look at Chinese and Japanese Data. J. Geophys. Res., 92: 3580–3596.

Li, C.Y. and Tang, Y.Q., 1983. Some problems on subdivision of Palaeo-plates in Asia. Acta Geol. Sinica, 1: 1–10 (in Chinese with English abstract).

Li, C.Y., Wang, Q., Liu, X.Y. and Tang, Y.Q., 1984. Tectonic evolution of Asia. Bull. Chinese Acad. Geol. Sci., 10: 3–11 (in Chinese with English abstract).

Li, G., 1986. Preliminary study on the paleosuture zone between the Sino-Korea and Siberia plates in east Inner Mongolia. KeXue Tongbao, 14: 1093–1096 (in Chinese).

Li, Y., 1990. An apparent polar wander path from the Tarim block, China. In: M. Kono and B.C. Burchfiel (Editors), Tectonics of Eastern Asia and Western Pacific Continental Margin. Tectonophysics, 182 (this volume): 31–41.

Li, Y., McWilliams, M., Cox, A., Sharps, R., Li, Y.A., Gao, Z., Zhang, Z. and Zhai, Y., 1988. Late Permian paleomagnetic

pole from dikes of the Tarim craton, China. Geology, 16: 275–278.

Lin, J.L., 1984. The apparent polar wander paths for the North and South China blocks. Ph.D. Thesis, University of California, Santa Barbara, Calif., 248 pp.

Lin, J.L., 1989. Advances in the paleomagnetic study of China. In: S.J. Wang (Editor), Advances in Geosciences, 1. Contributions to the International Geological Conference. China Ocean Press, Beijing, pp. 20–33.

Liu, H.S., 1987. Preliminary results of paleomagnetic study of Permian rocks in Inner Mongolia. 3rd Natl. Symp. on Paleomagnetism, Guangzhou, Abstr. (in Chinese).

Liu, J.Q., 1988. The Cenozoic volcanic episodes in northeast China. Acta Petrol. Sinica, 1: 1–10 (in Chinese with English abstract).

Liu, X.H. and Hao, J., 1989. Structure and tectonic evolution of the Tongbai–Dabie range in the east Qinling collisional belt, China. Tectonics, 8: 637–645.

Maruyama, S., Li, J.G. and Seno, T., 1989. Mesozoic and Cenozoic Evolution of Asia. In: Z. Ben-Avraham (Editor), The Evolution of the Pacific Ocean Margins. Oxford Monogr. Geol. Geophys., 8: 75–99.

Mattauer, M., Matte, Ph., Malevicille, J., Tapponnier, P., Maluski, H., Xu, Z.Q., Lu, Y.L. and Tang, Y.Q., 1985. Tectonics of the Qinling Belt: build-up and evolution of eastern Asia. Nature, 327: 496–500.

McElhinny, M.W., Embleton, B.J., Ma, X.H. and Zhang, Z.K., 1981. Fragmentation of Asia in the Permian. Nature, 293: 212–215.

McFadden, P.L. and McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in palaeomagnetism. Earth Planet. Sci. Lett., 87: 161–172.

McFadden, P.L., Ma, X.H., McElhinny, M.W. and Zhang, Z.K., 1988. Permo-Triassic magnetostratigraphy in China: northern Tarim. Earth Planet. Sci. Lett., 87: 152–160.

Molnar, P. and Deng, Q., 1984. Faulting associated with large earthquakes and the average rate of deformation in central and eastern Asia. J. Geophys. Res., 89: 6203–6227.

Mossakovskiy, A.A. and Dergunov, A.B., 1983. Caledonides of Kazakhstan and Central Asia: structure evolution and paleotectonic environments. Geotectonics, 17: 93–104.

Nalivkin, D.V., 1973. Geology of the U.S.S.R. University of Toronto Press, Toronto, Ont., 855 pp.

Nei Monggol Regional Stratigraphy Tables Compilation Team, 1978. Regional Stratigraphic Tables of Northern China, Nei Monggol Volume. Geological Publishing House, Beijing, 338 pp. (in Chinese).

Ni, J., 1978. Contemporary tectonics in the Tien Shan region, Earth Planet. Sci. Lett., 41: 347–354.

Nie, S.Y., Rowley, D.B. and Ziegler, A.M., 1989. Constraints on the locations of Asian microcontinents in Paleo-Tethys during the Late Palaeozoic. In: McKerrow and Scotese (Editors), Symposium Volume, Oxford Meeting (in press).

Opdyke, N.D., 1986. The Kiaman superchron in North America. EOS, Trans. Am. Geophys. Union, 67: 269 (abstract).

Opdyke, N.D., Huang, K., Xu, G., Zhang, W.Y. and Kent, D.V., 1986. Paleomagnetic results from the Triassic of the Yangtze platform. J. Geophys. Res., 91: 9553–9568.

Palmer, A.R., 1983. The decade of North American geology: 1983 Geologic time scale. Geology, 11: 503–504.

Piper, J.D.A., 1988. Paleomagnetic Data Base. Open University Press, Milton Keynes, 264 pp.

Pruner, P., 1987. Palaeomagnetism and palaeogeography of Mongolia in the Cretaceous, Permian and Carboniferous—preliminary data. Tectonophysics, 139: 155–167.

Quan, H., 1983. Granite types and related tin ore in Huanggang district, eastern Nei Monggol. Bull. Shengyang Inst. Geol. Mineral. Res., Chinese Acad. Geol. Sci., 6: 95–114 (in Chinese with English abstract).

Ren, J.S., Jiang, C.F., Zhang, Z.K. and Qin, D.Y., 1980. The Geotectonic Evolution of China. Science Press, Beijing, 124 pp. (in Chinese).

Ren, J.S., Jiang, C.F., Zhang, Z.K. and Qin, D.Y., 1987. The Geotectonic Evolution of China. Science Press/Springer-Verlag, Berlin, 203 pp.

Schmidt, P.W., Embleton, B.J.J. and Palmer, H.C., 1987. Pre- and post-folding magnetizations from the early Devonian Snowy River Volcanics and Buchan Caves Limestone, Victoria. Geophys. J. R. Astron. Soc., 91: 155–170.

Sengör, A.M.C., 1985. East Asian tectonic collage. Nature, 318: 16–17.

Sharps, R., Li, Y.P., McWilliams, M., Li, Y.G., Li, Q. and McKnight, C., 1988. Age of folding along the northwestern margin of the Tarim craton: evidence from remagnetized Late Paleozoic strata. EOS, Trans. Am. Geophys. Union, 69: 1171 (abstract).

Sharps, R., McWilliams, M., Li, Y.P., Cox, A., Zhang, Z.K., Zhai, Y.J., Gao, Z.J., Li, Y.A. and Li, Q., 1989. Lower Permian paleomagnetism of the Tarim Block, northwestern China. Earth Plant. Sci. Lett., 92: 275–291.

Snelling, N.J., 1987. Measurement of geological time and the geological time scale. Modern Geol., 11: 365–374.

Tamrazyan, G.P., 1971. Siberian continental drift. Tectonophysics, 11: 433–460.

Tan, X.D., 1988. Paleomagnetism of Mesozoic rock units from Shanxi–Shaanxi region and the tectonic implications. Master Thesis, Zhejiang University, Hangzhou, 166 pp. (in Chinese with English abstract).

Van der Voo, R., 1989. Paleomagnetism of North America: the craton, its margins, and the Appalachian belt. In: L.C. Pakiser and W.D. Mooney (Editors), Geophysical Framework of the Continental United States, Geol. Soc. Am., Mem. (in press).

Van der Voo, R. and French, R.B., 1974. Apparent polar wandering path for the Atlantic bordering continents: Late Carboniferous to Eocene. Earth-Sci. Rev., 10: 99–119.

Wang, H.Z., 1981. Geotectonic units of China from the viewpoint of mobilism. Earth Sci. J. Wuhan College Geol., 1: 42–66. (in Chinese).

Wang, Q., 1986. Recognition of suture line between the Sino-korean and Siberian paleoplates in the middle part of Inner

Mongolia, China Acta Geol. Sinica, 1: 31–43 (in Chinese with English abstract).

Wang, Q. and Liu, X.Y., 1986. Paleoplate tectonics between Cathaysia and Angaraland in Inner Mongolia of China. Tectonics, 5: 1073–1088.

Wang, T.D., 1986. Isotopic ages of the Upper Jurassic–Cretaceous volcanic rocks from the Hulun Buir League and the Ju Ud League, Inner Mongolia. Geol. Rev., 32: 526–531 (in Chinese with English abstract).

Waston, M.P., Hayward, A.B., Parkinson, D.M. and Zhang, Z.M., 1987. Plate tectonic history basin development and petroleum source rock deposition onshore China. Mar. Pet. Geol., 4: 205–225.

Westphal, M., Bazhenov, M., Lauer, J., Pechersky, M. and Sibuet, J., 1986. Paleomagnetic implications of the evolution of the Tethys belt from the Atlantic Ocean to the Pamirs since the Triassic. Tectonophysics, 123: 37–82.

Wu, H.N., 1988. Apparent polar wander paths and paleolatitude distributions for the North and South China blocks, the geotectonic evolution of Qinling belt. Ph.D. Thesis, Institute of Geology, Academia Sinica, Beijing, 176 pp. (in Chinese with English abstract).

Wu, S.Z., 1983. Preliminary study on the Permian phytogeography of Xinjiang in Chinese). In: Paleobiogeographic Provinces of China. Science Press, Beijing, pp. 91–99.

Xinjiang Regional Stratigraphy Tables Compilation Team, 1981. Regional Stratigraphic Tables of Northwestern China, Xinjiang Volume. Geological Publishing House, Beijing, 496 pp. (in Chinese).

Xiong, Y.S., Wu, Z.S. and Luo, C.C., 1987. Paleomagnetic implication on correlation and division of Cretaceous strata in north margin of the North China Block. 3rd Natl. Symp. on Paleomagnetism, Guangzhou, Abstr. (in Chinese).

Yang, S. and Yang, W., 1985. Geotectonics of China. Geological Publishing House, Beijing, 341 pp. (in Chinese).

Yang, Z.Y., Cheng, Y.Q. and Wang, H.Z., 1986. The Geology of China, Clarendon Press, Oxford, 303 pp.

Yermolov, P.V., Patalakha, Y.I., Yefimov, I.A., Polyanskiy, N.V., Ponomareva, A.P., Naydenov, B.M. and Khokhlov, P.I., 1984. Metamorphic complexes and tectonic features of the Zaysan fold belt and Rudnyy Altai. Geotectonics, 18: 334–343.

Zhai, Y., Li, Y., McWilliams, M., Cox, A., Sharps, R., Zhang, Z., Li, Y.A. and Li, Q., 1987. Late Carboniferous paleomagnetism of the Tarim craton, China. EOS, Trans. Am. Geophys. Union, 68: 1259 (abstract).

Zhai, Y., Zhang, Z., Li, Y.A., Li, Q., Li, Y., McWilliams, M., Cox, A. and Sharps, R., 1988. A study of Upper Carboniferous paleomagnetism of the Tarim Block. Geosciences, 2: 43–56 (in Chinese with English abstract).

Zhang, Z.K., 1984. Sino-Korean blocks and Yangtze block as part of the Pacifica continent in the late Paleozoic. Bull. Chinese Acad. Geol. Sci., 9: 45–54 (in Chinese with English abstract).

Zhang, Zh.M., Liou, J.G. and Coleman, R.G., 1984. An outline of plate tectonics of China. Geol. Soc. Am. Bull., 95: 295–312.

Zhang, Z.M., Liou, J.G. and Coleman, R.G., 1989. The Mesozoic and Cenozoic tectonism of eastern China. In: Z. Ben-Avraham (Editor), The Evolution of the Pacific Ocean Margins. Oxford Monogr. Geol. Geophys., 8: 124–139.

Zhao, X. and Coe, R.S., 1987. Palaeomagnetic constraints on the collision and rotation of North and South China. Nature, 327: 141–144.

Zhao, X. and Coe, R., 1989. Tectonic implications of Permo-Triassic paleomagnetic results from North and South China. In: J.W. Hillhouse (Editor), Deep Structure and Past Kinematics of Accreted Terranes. IUG-G Geophys. Monogr., 5: 267–283.

Zhao, X., Coe, R., Wu, H. and Zhang, C., 1988. Evidence for Remagnetization of Paleozoic Rock Units From Junggar Basin of Xinjiang Province, China. EOS, Trans. Am. Geophys. Union, 69: 1170 (abstract).

Zhao, Z.C., Zhou, L.G., Huang, T.B. and Zhang, J.S., 1983. The features of the Lhaba-barbut arcuate rift belt. Bull. Xi'an Inst. Geol. Mineral. Res., Chinese Acad. Geol. Sci., 6: 41–49 (in Chinese with English abstract).

Zheng, Z., Kono, M., Tsunakawa, H., Zhu, X.Y. and Hao, T., 1989. An apparent polar wander path for North China block since middle Jurassic. In: M. Kono (Editor), Northeast Asian Tectonics, 3. Tokyo Institute of Technology, Tokyo, pp. 22–41.

Zhu, B.Q., 1984. Ancient oceanic crust and plate tectonics in southwestern section of western Junggar. Bull. Chinese Acad. Geol. Sci., 10: 137–150 (in Chinese with English abstract).

Zonenshain, L.P., 1973. The evolution of central Asiatic geosynclines through seafloor spreading. Tectonophysics, 19: 212–232.

Zonenshain, L.P., Kuzmin, M.I. and Kononov, M.V., 1985. Absolute reconstructions of the Paleozoic oceans. Earth Planet. Sci. Lett., 74: 103–116.